

Question ID 3409707e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 3409707e

$$\begin{aligned}x + y &= 125 \\ x + y + y &= 155\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of y ?

ID: 3409707e Answer

Correct Answer:
30

Rationale

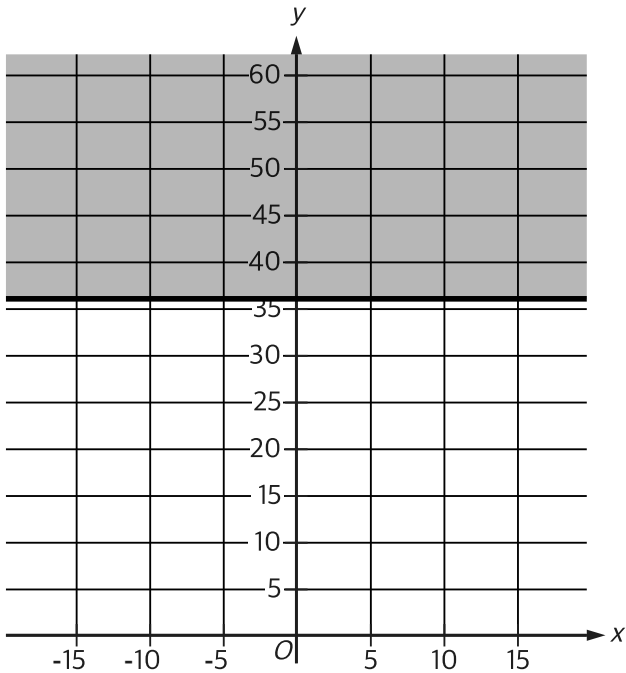
The correct answer is **30**. It’s given that $x + y = 125$. Substituting **125** for $x + y$ in the second equation of the given system yields $125 + y = 155$. Subtracting **125** from both sides of this equation yields $y = 30$. Therefore, the value of y is **30**.

Question Difficulty:
Easy

Question ID 744ee7d7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 744ee7d7



The shaded region shown in the graph represents all the solutions to which inequality?

- A. $x \leq 36$
- B. $x \geq 36$
- C. $y \leq 36$
- D. $y \geq 36$

ID: 744ee7d7 Answer

Correct Answer:

D

Rationale

Choice D is correct. The equation for the line representing the boundary of the shaded region can be written in the form $y = b$, where $(0, b)$ is the y -intercept of the line in the xy -plane. For the graph shown in the xy -plane, the boundary line passes through the points $(-15, 36)$ and $(0, 36)$. Thus, the equation of the boundary line is $y = 36$. Since the shaded region represents all the points on and above this boundary line, it follows that the graph represents all the solutions to the inequality $y \geq 36$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

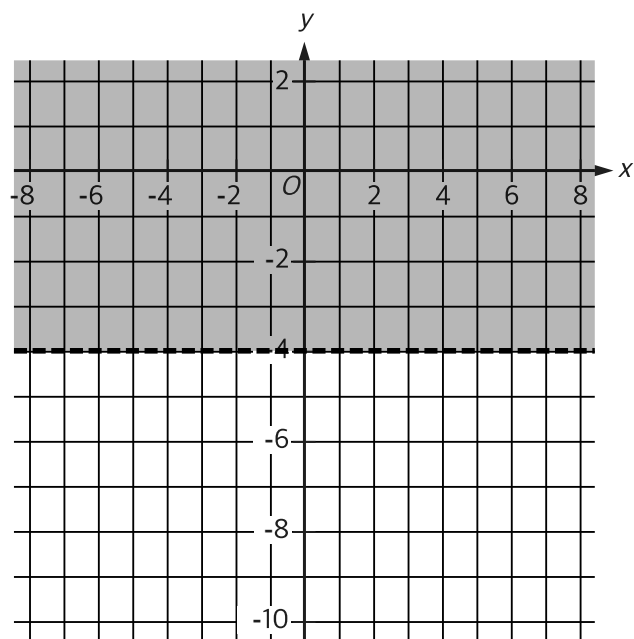
Question Difficulty:

Easy

Question ID 187f74bc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 187f74bc



The shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. What is the value of r ?

- A. 15
- B. 4
- C. -4
- D. -15

ID: 187f74bc Answer

Correct Answer:

D

Rationale

Choice D is correct. The line representing the boundary of the shaded region shown is a horizontal line; therefore, the equation of the line can be written as $y = -4$. Since the shaded region represents all of the points above this boundary line, it follows that the shaded region represents the solutions to the inequality $y > -4$. It's given that the shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. The inequality $y > -4$ can be rewritten in the form $ry < 60$ by multiplying both sides of the inequality by -15 , which yields $(-15)y < 60$. Therefore, the value of r is -15 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

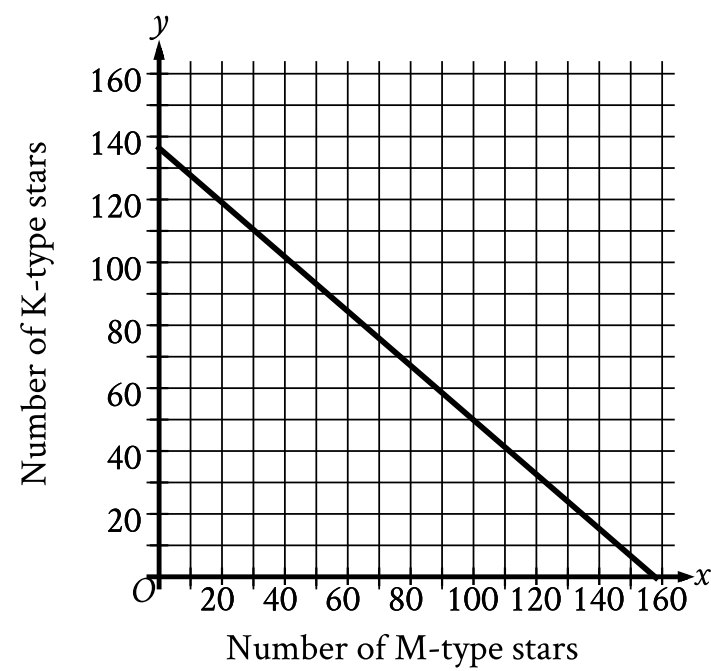
Question Difficulty:

Hard

Question ID 1a1a95de

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: 1a1a95de



A certain open star cluster contains M-type stars and K-type stars. The estimated total mass of M-type and K-type stars in this open star cluster is **127,882 quettagrams**. The graph shown models the possible combinations of the number of M-type stars, x , and K-type stars, y , that could be in this open star cluster if all the M-type stars have the same estimated mass and all the K-type stars have the same estimated mass. Based on the graph, which of the following is closest to the estimated mass, in quettagrams, of each M-type star in this cluster?

- A. 811
- B. 938
- C. 51,904
- D. 75,978

ID: 1a1a95de Answer

Correct Answer:
A

Rationale

Choice A is correct. The x -intercept of the graph shown represents the situation in which the cluster contains only M-type stars. Based on the graph, the x -intercept is approximately **(158, 0)**. Therefore, if there are **0** K-type stars in the cluster, then there are approximately **158** M-type stars in the cluster. It's given that the total mass of the stars in this cluster is **127,882** quettagrams. Therefore, the estimated mass of each M-type star is approximately $\frac{127,882}{158}$ quettagrams, or approximately **809.38** quettagrams. Thus, of the given choices, **811** is the closest to the estimated mass, in quettagrams, of each M-type star in this cluster.

Choice B is incorrect. This is the value closest to the estimated mass, in quettagrams, of each K-type star, not each M-type star, in this cluster.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID 465c73ad

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 465c73ad

$10x = 86$

What value of x is the solution to the given equation?

ID: 465c73ad Answer

Correct Answer:

8.6, 43/5

Rationale

The correct answer is **8.6**. Dividing both sides of the given equation by **10** yields $x = 8.6$. Therefore, **8.6** is the solution to the given equation. Note that 8.6 and 43/5 are examples of ways to enter a correct answer.

Question Difficulty:

Easy

Question ID 3dffe9d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 3dffe9d2

Josie purchased a pass that cost **\$19.50** to tour a nature preserve. She also purchased **10** identical stickers to give to her nieces and nephews. The pass and the stickers cost **\$62.00** total. What was the cost of one sticker?

- A. **\$1.95**
- B. **\$4.25**
- C. **\$6.20**
- D. **\$8.15**

ID: 3dffe9d2 Answer

Correct Answer:

B

Rationale

Choice B is correct. It's given that Josie purchased a pass to tour a nature preserve and **10** identical stickers for **\$62.00** total. It's also given that the pass cost **\$19.50**. Let x represent the cost, in dollars, of one sticker. This relationship can be represented by the equation $19.50 + 10x = 62$. Subtracting **19.50** from both sides of this equation yields $10x = 42.50$. Dividing both sides of this equation by **10** yields $x = 4.25$. Therefore, the cost of one sticker was **\$4.25**.

Choice A is incorrect. This is the cost of one sticker if Josie purchased **10** identical stickers for **\$19.50**, not a pass and **10** identical stickers for **\$62.00**.

Choice C is incorrect. This is the cost of one sticker if Josie purchased **10** identical stickers for **\$62.00**, not a pass and **10** identical stickers for **\$62.00**.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID 4edecdba

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: 4edecdba

$8x + 11y = 170$

The equation gives the possible combinations of the number of **2009** premium grade Log Cabin Pennies, x , and the number of **1996** select grade Lincoln Pennies, y , in a collection that is worth a total of **\$170**. If there are **6** **1996** select grade Lincoln Pennies in the collection, how many **2009** premium grade Log Cabin Pennies are in the collection?

ID: 4edecdba Answer

Correct Answer:
13

Rationale

The correct answer is **13**. It's given that the equation $8x + 11y = 170$ gives the possible combinations of the number of **2009** premium grade Log Cabin Pennies, x , and the number of **1996** select grade Lincoln Pennies, y , in a collection that is worth a total of **\$170**. It's also given that there are **6** **1996** select grade Lincoln Pennies in the collection, and thus $y = 6$. Substituting **6** for y in the given equation yields $8x + 11(6) = 170$, or $8x + 66 = 170$. Subtracting **66** from both sides of this equation yields $8x = 104$. Dividing both sides of this equation by **8** yields $x = 13$. Therefore, there are **13** **2009** premium grade Log Cabin Pennies in the collection.

Question Difficulty:
Easy

Question ID 00ec9102

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 00ec9102

Leo goes to a packing store to buy containers and tape. Leo has **\$15**. Each container costs **\$1.87** and each roll of tape costs **\$2.40**. Which inequality represents the relationship between the number of containers, ***c***, and the number of rolls of tape, ***t***, Leo can buy?

- A. $1.87c + 2.40t \leq 15$
- B. $1.87c + 2.40t \geq 15$
- C. $2.40c + 1.87t \leq 15$
- D. $2.40c + 1.87t \geq 15$

ID: 00ec9102 Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that each container costs **\$1.87** and ***c*** represents the number of containers. It follows that **$1.87c$** represents the cost, in dollars, of ***c*** containers. It's also given that each roll of tape costs **\$2.40** and ***t*** represents the number of rolls of tape. It follows that **$2.40t$** represents the cost, in dollars, of ***t*** rolls of tape. Therefore, the total cost, in dollars, for ***c*** containers and ***t*** rolls of tape is **$1.87c + 2.40t$** . It's given that Leo has **\$15**. It follows that the total cost, in dollars, must be less than or equal to **15**. Therefore, the inequality **$1.87c + 2.40t \leq 15$** represents this relationship.

Choice B is incorrect. This inequality represents a relationship where Leo has at least **\$15**, not at most **\$15**.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID 4047cfa7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	

ID: 4047cfa7

For a literature class, a student's essay needs to have no more than **5,000** words. If a student's essay already has **3,026** words, what is the maximum number of additional words the student can include in the essay?

- A. **1,974**
- B. **3,026**
- C. **5,000**
- D. **8,026**

ID: 4047cfa7 Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that a student's essay needs to have no more than **5,000** words. It's also given that the student's essay already has **3,026** words. Let x represent the number of additional words the student includes in the essay. After including x additional words, the essay will have a total of $3,026 + x$ words. Since the essay needs to have no more than **5,000** words, it follows that $3,026 + x \leq 5,000$. Subtracting **3,026** from both sides of this inequality yields $x \leq 1,974$. Thus, the maximum number of additional words the student can include in the essay is **1,974**.

Choice B is incorrect. This is the number of words already in the student's essay, rather than the maximum number of additional words the student can include.

Choice C is incorrect. This is the total number of words the essay can contain, rather than the maximum number of additional words the student can include.

Choice D is incorrect. This is the number of words the essay would contain if the student added **5,000** more words, rather than the maximum number of additional words the student can include.

Question Difficulty:

Easy

Question ID 343aaff6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 343aaff6

For the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 7 and passes through the point $(0, 5)$. Which equation defines f ?

- A. $f(x) = 5x$
- B. $f(x) = 35x$
- C. $f(x) = 7x + 5$
- D. $f(x) = 12x + 5$

ID: 343aaff6 Answer

Correct Answer:
C

Rationale

Choice C is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m is the slope of the graph of $y = f(x)$ in the xy -plane and $(0, b)$ is the y -intercept of the graph. It's given that for the linear function f , the graph of $y = f(x)$ in the xy -plane has a slope of 7 . Therefore, $m = 7$. It's also given that the graph of $y = f(x)$ in the xy -plane passes through the point $(0, 5)$. Therefore, the y -intercept of the graph is $(0, 5)$, and it follows that $b = 5$. Substituting 7 for m and 5 for b in the equation $f(x) = mx + b$ yields $f(x) = 7x + 5$.

Choice A is incorrect. This equation defines a function whose graph has a slope of 5 , not 7 , and passes through the point $(0, 0)$, not $(0, 5)$.

Choice B is incorrect. This equation defines a function whose graph has a slope of 35 , not 7 , and passes through the point $(0, 0)$, not $(0, 5)$.

Choice D is incorrect. This equation defines a function whose graph has a slope of 12 , not 7 .

Question Difficulty:
Easy

Question ID 346d5c2e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 346d5c2e

The function f is defined by $f(x) = 9(2x + 3)$. For what value of x does $f(x) = 63$?

- A. 2
- B. 5
- C. 7
- D. 30

ID: 346d5c2e Answer

Correct Answer:
A

Rationale

Choice A is correct. Substituting **63** for $f(x)$ in the given function yields $63 = 9(2x + 3)$. Dividing both sides of this equation by **9** yields $7 = 2x + 3$. Subtracting **3** from both sides of this equation yields $4 = 2x$. Dividing both sides of this equation by **2** yields $2 = x$. Therefore, $f(x) = 63$ when the value of x is **2**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 68f343c9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 68f343c9

If $f(x) = x + 6$ and $g(x) = 6x$, what is the value of $7f(2) - g(2)$?

- A. ~~−4~~
- B. 8
- C. ~~42~~
- D. ~~44~~

ID: 68f343c9 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that $f(x) = x + 6$. Substituting **2** for x in the function f yields $f(2) = 2 + 6$, or $f(2) = 8$. It's also given that $g(x) = 6x$. Substituting **2** for x in the function g yields $g(2) = (6)(2)$, or $g(2) = 12$. Substituting **8** for $f(2)$ and **12** for $g(2)$ in the expression $7f(2) - g(2)$ yields $7(8) - 12$, or **44**. Therefore, the value of $7f(2) - g(2)$ is **44**.

Choice A is incorrect. This is the value of $f(2) - g(2)$, not $7f(2) - g(2)$.

Choice B is incorrect. This is the value of $f(2)$, not $7f(2) - g(2)$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 5ab9903f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: 5ab9903f

The cost to rent a bus from Company X is **\$950** for the first **3** hours and an additional **\$50** per hour for each hour after the first **3** hours. If the total cost to rent the bus from Company X for t hours, where $t > 3$, is **\$1,150**, which equation represents this situation?

- A. $950(t - 3) + 50t = 1,150$
- B. $950(3t) + 50t = 1,150$
- C. $950 + 50(t - 3) = 1,150$
- D. $950 + 50(3t) = 1,150$

ID: 5ab9903f Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that the cost to rent a bus is **\$950** for the first **3** hours and an additional **\$50** per hour for each hour after the first **3** hours. It's also given that t represents the total number of hours and $t > 3$. Therefore, the number of additional hours after the first **3** hours can be represented by the expression $t - 3$. The cost for these additional hours is **\$50** per hour, so the cost for the additional hours can be represented by the expression $50(t - 3)$. The total cost can be calculated by adding the cost for the first **3** hours to the cost for the additional hours and can be represented by the expression $950 + 50(t - 3)$. It's also given that the total cost to rent the bus from the company for t hours is **\$1,150**. Thus, the equation that represents this situation is $950 + 50(t - 3) = 1,150$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty:
Medium

Question ID 63d03c0b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 63d03c0b

For a particular car, the linear function f gives the predicted power, in brake horsepower (**bhp**), for engine speeds between **1,000 revolutions per minute (rpm)** and **6,000 rpm**. According to this function, the car's predicted power is **433 bhp** at an engine speed of **3,331 rpm** and **600 bhp** at an engine speed of **4,500 rpm**. The equation $f(x) = \frac{1}{7}(x - a) + 433$ defines f , where x is the engine speed, **in rpm**, and a is a constant. What is the value of a ?

ID: 63d03c0b Answer

Correct Answer:
3331

Rationale

The correct answer is **3,331**. For engine speeds between **1,000 revolutions per minute (rpm)** and **6,000 rpm**, it's given that the equation $f(x) = \frac{1}{7}(x - a) + 433$ defines the linear function f and gives the predicted power, in brake horsepower (**bhp**), at an engine speed of x **rpm**, where a is a constant. It's also given that the car's predicted power is **433 bhp** at an engine speed of **3,331 rpm**. Substituting **3,331** for x and **433** for $f(x)$ in the equation yields $433 = \frac{1}{7}(3,331 - a) + 433$. Subtracting **433** from both sides of this equation yields $0 = \frac{1}{7}(3,331 - a)$. Multiplying both sides of this equation by **7** yields $0 = 3,331 - a$. Adding a to both sides of this equation yields $a = 3,331$. Thus, the value of a is **3,331**.

Question Difficulty:
Hard

Question ID 58477a6c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: 58477a6c

$2x + 9y = 7$

The given equation is one equation in a system of two linear equations. If the system of equations has at least one solution, which of the following equations could be the other equation in the system?

- I. $3x + 13.5y = 10.5$
- II. $3x - 13.5y = 10.5$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

ID: 58477a6c Answer

Correct Answer:
C

Rationale

Choice C is correct. A system of two linear equations in two variables has at least one solution if the equations represent the same line or lines that intersect at exactly one point. Statement I gives the equation $3x + 13.5y = 10.5$. Multiplying both sides of the given equation, $2x + 9y = 7$, by $\frac{3}{2}$ yields $3x + 13.5y = 10.5$, so the equation in statement I is equivalent to the given equation. Therefore, the system formed by these two equations represents the same line and has infinitely many solutions. Statement II gives the equation $3x - 13.5y = 10.5$. Adding the right- and left-hand sides of $3x + 13.5y = 10.5$ and $3x - 13.5y = 10.5$ yields $3x + 13.5y + 3x - 13.5y = 10.5 + 10.5$, or $6x = 21$. Dividing both sides of this equation by 6 yields $x = \frac{21}{6}$. Therefore, the lines represented by these equations intersect at exactly one point, where $x = \frac{21}{6}$, and the system formed by the given equation and the equation in statement II has one solution. It follows that both the equation in statement I and the equation in statement II could be the other equation in a system of equations with at least one solution.

Choice A is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

Choice B is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

Choice D is incorrect. Both the equation in statement I and the equation in statement II could be the other equation in this system of equations with at least one solution.

Question Difficulty:
Hard

Question ID deee9063

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: deee9063

Line k is defined by $y = 6x + 4$. Line j is parallel to line k in the xy -plane and passes through the point $(0, 5)$. Which equation defines line j ?

- A. $y = 6x + 5$
- B. $y = -5x + 5$
- C. $y = -6x + 5$
- D. $y = 5x + 5$

ID: deee9063 Answer

Correct Answer:
A

Rationale

Choice A is correct. An equation that defines a line in the xy -plane can be written as $y = mx + b$, where m is the slope of the line and $(0, b)$ is its y -intercept. It's given that line k is defined by the equation $y = 6x + 4$; therefore, the slope of line k is 6 . Since line j is parallel to line k in the xy -plane and parallel lines have equal slopes, it follows that the slope of line j is 6 . It's also given that line j passes through the point $(0, 5)$, which is its y -intercept. Substituting 6 for m and 5 for b in the equation $y = mx + b$ yields $y = 6x + 5$. Therefore, the equation $y = 6x + 5$ defines line j .

Choice B is incorrect. This equation defines a line that has a slope of -5 , not 6 .

Choice C is incorrect. This equation defines a line that has a slope of -6 , not 6 .

Choice D is incorrect. This equation defines a line that has a slope of 5 , not 6 .

Question Difficulty:
Easy

Question ID 23c5d458

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	

ID: 23c5d458

x	$g(x)$
1	32
2	28
3	24
4	20

For the linear function g , the table shows four values of x and their corresponding values of $g(x)$. The function can be written as $g(x) = mx + b$, where m and b are constants. What is the value of b ?

- A. 4
- B. 16
- C. 32
- D. 36

ID: 23c5d458 Answer

Correct Answer:
D

Rationale

Choice D is correct. It's given that for the linear function g , the table shows four values of x and their corresponding values of $g(x)$. It's also given that the function can be written as $g(x) = mx + b$, where m and b are constants. The table shows that when the value of x is 1, the corresponding value of $g(x)$ is 32. Substituting 1 for x and 32 for $g(x)$ in $g(x) = mx + b$ yields $32 = m(1) + b$, or $32 = m + b$. Subtracting b from both sides of this equation yields $32 - b = m$. The table also shows that when the value of x is 2, the corresponding value of $g(x)$ is 28. Substituting 2 for x and 28 for $g(x)$ in $g(x) = mx + b$ yields $28 = m(2) + b$, or $28 = 2m + b$. Substituting $32 - b$ for m in this equation yields $28 = 2(32 - b) + b$. Applying the distributive property to the right-hand side of this equation yields $28 = 64 - 2b + b$, or $28 = 64 - b$. Subtracting 64 from both sides of this equation yields $-36 = -b$. Dividing both sides of this equation by -1 yields $36 = b$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID d76ab4f5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	

ID: d76ab4f5

$4x + 1 = 33$

Which equation has the same solution as the given equation?

- A. $4x = 32$
- B. $4x = 5$
- C. $4x = 1$
- D. $4x = -32$

ID: d76ab4f5 Answer

Correct Answer:

A

Rationale

Choice A is correct. Subtracting 1 from both sides of the given equation, $4x + 1 = 33$, yields $4x = 32$. Therefore, $4x = 32$ has the same solution as the given equation.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID 820ad8e8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	

ID: 820ad8e8

In a chess tournament, each participant earns **1** point for each game the participant plays that ends in a draw and **3** points for each game the participant wins. A certain participant in this tournament has earned **41** points. Which equation represents this situation, where *d* represents the number of games this participant has played that ended in a draw and *w* represents the number of games this participant has won?

- A. $d + 3w = 41$
- B. $3d + w = 41$
- C. $d + \frac{w}{3} = 41$
- D. $\frac{d}{3} + w = 41$

ID: 820ad8e8 Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that each participant earns **1** point for each game the participant plays that ends in a draw and **3** points for each game the participant wins. Since *d* represents the number of games that a certain participant has played that ended in a draw and *w* represents the number of games this participant has won, this participant has earned a total of **$1d + 3w$** , or **$d + 3w$** , points. Since this participant earned **41** points, the equation **$d + 3w = 41$** represents this situation.

Choice B is incorrect. This equation represents the situation where the participant earned **3** points for each game played that ended in a draw and **1** point for each game won and earned **41** points.

Choice C is incorrect. This equation represents the situation where the participant earned **1** point for each game played that ended in a draw and **$\frac{1}{3}$** point for each game won and earned **41** points.

Choice D is incorrect. This equation represents the situation where the participant earned **$\frac{1}{3}$** point for each game played that ended in a draw and **1** point for each game won and earned **41** points.

Question Difficulty:

Easy

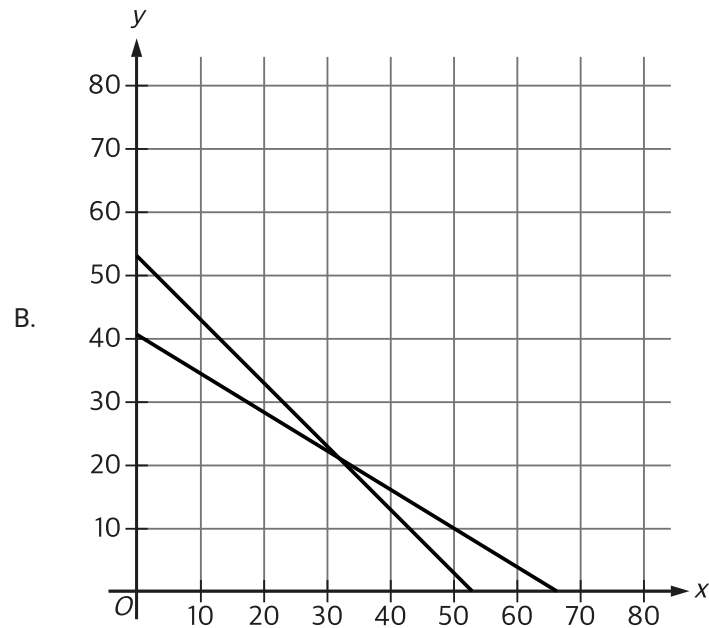
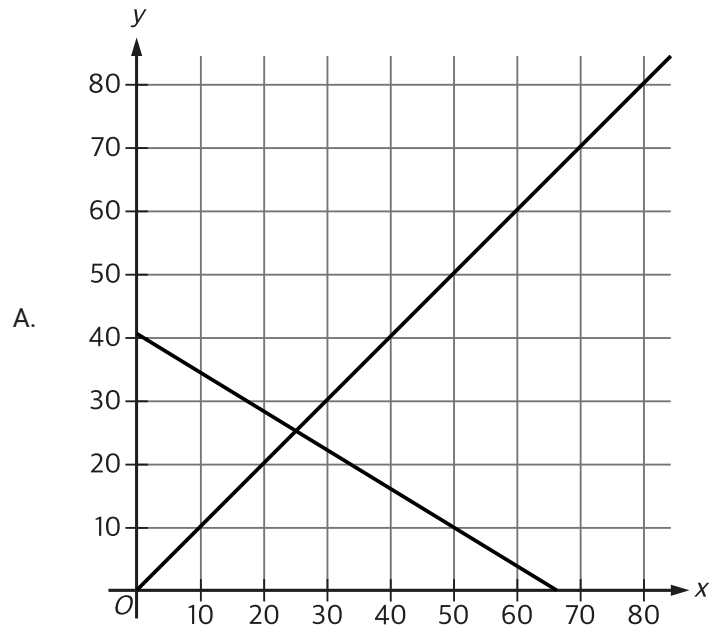
Question ID 36f068e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

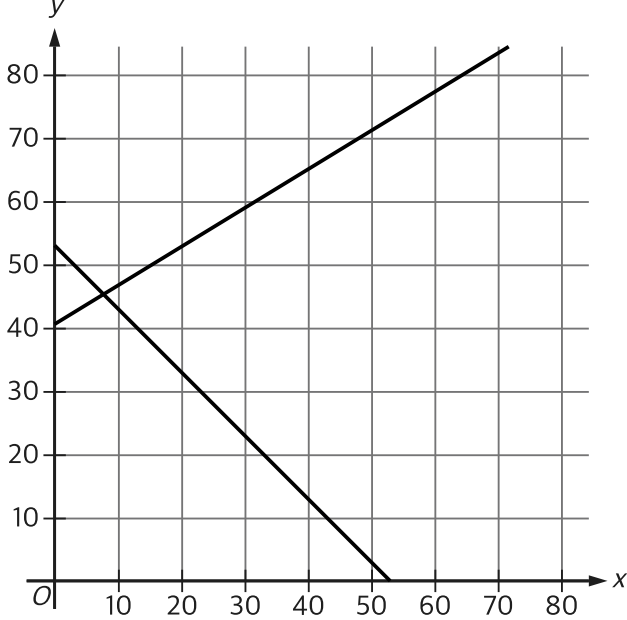
ID: 36f068e2

$$x + y = 53$$
$$11x + 18y = 730$$

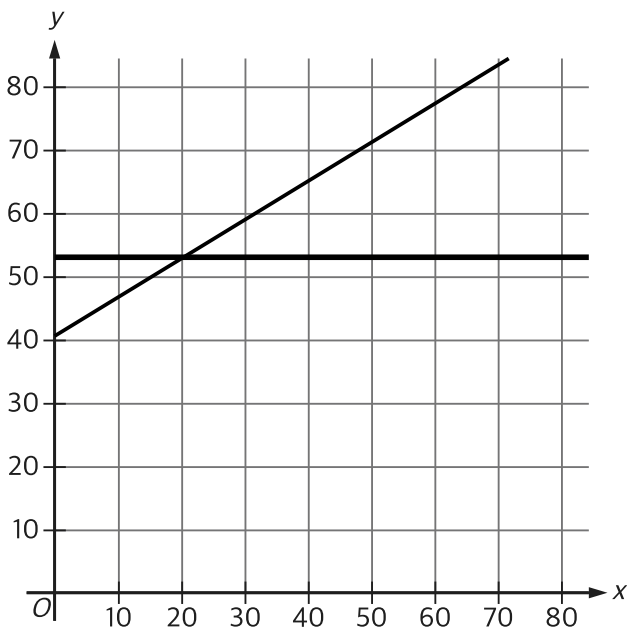
The given equations represent the possible numbers of beach chairs, x , and umbrellas, y , rented at a park last month and the total spent, in dollars, to rent those beach chairs and umbrellas. Which of the following graphs represents this situation?



C.



D.



ID: 36f068e2 Answer

Correct Answer:

B

Rationale

Choice B is correct. The intercepts of the graph of each equation can be found by substituting 0 for one variable at a time. For the equation $x + y = 53$, substituting 0 for y yields $x = 53$, so the x -intercept of the graph of the equation is $(53, 0)$. Substituting 0 for x in this equation yields $y = 53$, so the y -intercept of the graph of the equation is $(0, 53)$. Thus, the graph of the equation $x + y = 53$ has intercepts at $(53, 0)$ and $(0, 53)$. For the equation $11x + 18y = 730$, substituting 0 for y yields $x = \frac{730}{11}$, so the x -intercept of the graph of the equation is approximately $(66, 0)$. Substituting 0 for x in this equation yields $y = \frac{730}{18}$, so the y -intercept of the graph of the equation is approximately $(0, 41)$. Thus, the graph of the equation $11x + 18y = 730$ has intercepts at approximately $(66, 0)$ and $(0, 41)$. Choice B is a graph that shows one line with intercepts at $(0, 53)$ and $(53, 0)$ and another line with intercepts at approximately $(0, 41)$ and $(66, 0)$. Thus, choice B represents this situation.

Choice A is incorrect. This graph represents the equations $y = x$ and $11x + 18y = 730$.

Choice C is incorrect. This graph represents the equations $x + y = 53$ and $-11x + 18y = 730$.

Choice D is incorrect. This graph represents the equations $y = 53$ and $-11x + 18y = 730$.

Question Difficulty:

Medium

Question ID d1aeb6f1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	

ID: d1aeb6f1

$$10x = 110$$

$$6x - 63 = y$$

The solution to the given system of equations is (x, y) . What is the value of y ?

- A. 63
- B. 11
- C. 10
- D. 3

ID: d1aeb6f1 Answer

Correct Answer:

D

Rationale

Choice D is correct. Dividing both sides of the equation $10x = 110$ by 10 yields $x = 11$. Substituting 11 for x in the equation $6x - 63 = y$ yields $6(11) - 63 = y$, which is equivalent to $66 - 63 = y$, or $3 = y$. Therefore, the value of y is 3 .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x , not y .

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Easy

Question ID 253985c2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 253985c2

Which expression is equivalent to $6x + 5x + 4y$?

- A. $15x$
- B. $15y$
- C. $11x + 4y$
- D. $30x + 4y$

ID: 253985c2 Answer

Correct Answer:
C

Rationale

Choice C is correct. In the given expression, $6x$ and $5x$ are like terms. Combining these like terms yields $11x$. It follows that the expression $6x + 5x + 4y$ is equivalent to $11x + 4y$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID b0543157

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: b0543157

$$g(x) = \frac{x^2 - x - a}{x^3 - x - b}$$

The function g is defined by the given equation, where a and b are constants. In the xy -plane, the graph of $y = g(x)$ passes through the point $(0, 22)$, and $g(-22) = 0$. What is the value of b ?

- A. 23
- B. 22
- C. -22
- D. -23

ID: b0543157 Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that in the xy -plane, the graph of $y = g(x)$ passes through the point $(0, 22)$. It follows that $g(0) = 22$. Substituting 0 for x and 22 for $g(x)$ in the given equation yields $22 = \frac{0^2 - 0 - a}{0^3 - 0 - b}$, which is equivalent to $22 = \frac{-a}{-b}$, or $22 = \frac{a}{b}$. Multiplying each side of this equation by b yields $22b = a$. Substituting $22b$ for a in the equation defining the function g yields $g(x) = \frac{x^2 - x - 22b}{x^3 - x - b}$. It's given that $g(-22) = 0$. Substituting -22 for x and 0 for $g(x)$ in the equation $g(x) = \frac{x^2 - x - 22b}{x^3 - x - b}$ yields $0 = \frac{(-22)^2 - (-22) - 22b}{(-22)^3 - (-22) - b}$. Since a rational expression equals 0 if and only if its numerator is 0 , it follows that $(-22)^2 - (-22) - 22b = 0$, or $22^2 + 22 - 22b = 0$. Dividing each side of this equation by 22 yields $22 + 1 - b = 0$, or $23 - b = 0$. Adding b to each side of this equation yields $23 = b$, so the value of b is 23 .

Choice B is incorrect. This is the value of $g(0)$, not the value of b .

Choice C is incorrect. This is a value of x for which $g(x) = 0$, not the value of b .

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID ffdfe1e1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: ffdfe1e1

There are **240** players in a tennis competition that includes **4** rounds of matches. Each player in the competition will play a match against another player in round **1**. At the end of each round, the player who loses the match is eliminated and the player who won the match advances to the next round to play a match against another winning player. Which equation gives the number of players, p , eliminated at the end of round r , where $r \leq 4$?

- A. $p = 15\left(\frac{1}{2}\right)^r$
- B. $p = 15(2)^r$
- C. $p = 240\left(\frac{1}{2}\right)^r$
- D. $p = 240(2)^r$

ID: ffdfe1e1 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that one of the two players in a match is eliminated during each round. Therefore, the number of players decreases by half at the end of each round, and so the number of players eliminated at the end of a round also decreases by half. A decreasing exponential equation can be written in the form $p = a(1 - b)^r$, where b is the value of the decrease each round, r , and a is the value of p , the number of players, when $r = 0$. It's given that there are **240** players in the tennis competition. Therefore, $a = 240$. Since the number of players eliminated at the end of a round decreases by half, it follows that $b = \frac{1}{2}$. Substituting **240** for a and $\frac{1}{2}$ for b in the equation $p = a(1 - b)^r$ yields $p = 240\left(1 - \frac{1}{2}\right)^r$, or $p = 240\left(\frac{1}{2}\right)^r$.

Choice A is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which there are **15**, not **240**, players.

Choice B is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which there are **15**, not **240**, players, and for which the number of players doubles, rather than decreases by half, after each round.

Choice D is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which the number of players doubles, rather than decreases by half, after each round.

Question Difficulty:
Easy

Question ID 1e7a1deb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 1e7a1deb

If $a = 4k + 5r$ and $b = 7k - 12r + 3$, which expression is equivalent to $a - b$?

- A. $-3k + 17r + 3$
- B. $-3k + 17r - 3$
- C. $-3k - 7r - 3$
- D. $-3k - 7r + 3$

ID: 1e7a1deb Answer

Correct Answer:
B

Rationale

Choice B is correct. Substituting $4k + 5r$ for a and $7k - 12r + 3$ for b in the expression $a - b$ yields $(4k + 5r) - (7k - 12r + 3)$. This expression can be rewritten as $(4k + 5r) + (-1)(7k - 12r + 3)$. By applying the distributive property, this expression can be rewritten as $4k + 5r - 7k + 12r - 3$, which is equivalent to $(4k - 7k) + (5r + 12r) - 3$, or $-3k + 17r - 3$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

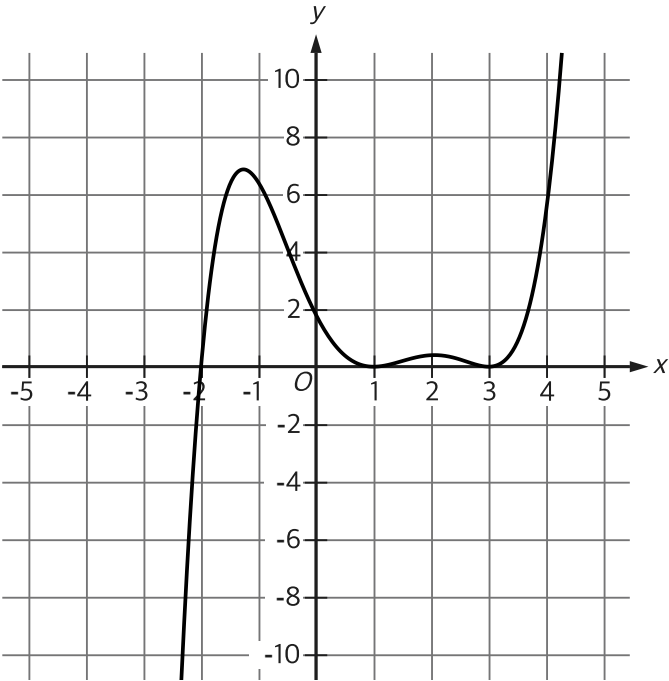
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID c946d5bd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: c946d5bd



The graph of $y = g(x)$ is shown. If $g(x + 28) = \frac{1}{10}(x - a)^2(x - b)^2(x - c)$, where a , b , and c are constants, which of the following could be the value of b ?

- A. 31
- B. 3
- C. -2
- D. -25

ID: c946d5bd Answer

Correct Answer:
D

Rationale

Choice D is correct. Substituting b for x in the given equation yields $g(b + 28) = \frac{1}{10}(b - a)^2(b - b)^2(b - c)$, which is equivalent to $g(b + 28) = \frac{1}{10}(b - a)^2(0)^2(b - c)$, or $g(b + 28) = 0$. It follows that when $x = b + 28$, $g(x) = 0$. There are three points where the graph of $y = g(x)$ intersects the x -axis: $(-2, 0)$, $(1, 0)$, and $(3, 0)$. Therefore, $g(-2) = 0$, $g(1) = 0$, and $g(3) = 0$. It follows that either $b + 28 = -2$, $b + 28 = 1$, or $b + 28 = 3$. Subtracting 28 from each side of the equation $b + 28 = -2$ yields $b = -30$. Subtracting 28 from each side of the equation $b + 28 = 1$ yields $b = -27$. Subtracting 28 from each side of the equation $b + 28 = 3$ yields $b = -25$. Thus, the value of b must be either -30 , -27 , or -25 . Of these values, only -25 is given as a choice.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This would be a possible value of b if the graph shown was the graph of $y = g(x + 28)$, not $y = g(x)$.

Choice C is incorrect. This would be the value of c , not the value of b , if the graph shown was the graph of $y = g(x + 28)$, not $y = g(x)$.

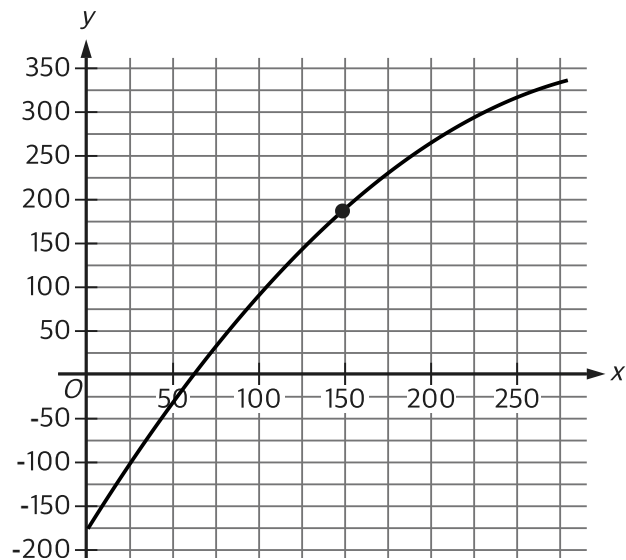
Question Difficulty:

Hard

Question ID 7d3a84bf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 7d3a84bf



The graph shows the estimated boiling point y , in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole, where $1 \leq x \leq 280$. Which statement is the best interpretation of the point $(149.02, 186.05)$?

- A. A normal paraffin with a molecular weight of **186.05** grams per mole has an estimated boiling point of **149.02** degrees Celsius.
- B. A normal paraffin with a molecular weight of **149.02** grams per mole has an estimated boiling point of **186.05** degrees Celsius.
- C. The minimum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of **149.02** grams per mole and an estimated boiling point of **186.05** degrees Celsius.
- D. The maximum estimated boiling point for normal paraffins corresponds to a paraffin with a molecular weight of **149.02** grams per mole and an estimated boiling point of **186.05** degrees Celsius.

ID: 7d3a84bf Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that the graph shows the estimated boiling point y , in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole. It follows that for the point $(149.02, 186.05)$, **149.02** represents the molecular weight, in grams per mole, of a normal paraffin and **186.05** represents its estimated boiling point, in degrees Celsius. Therefore, the best interpretation of the point $(149.02, 186.05)$ is that a normal paraffin with a molecular weight of **149.02** grams per mole has an estimated boiling point of **186.05** degrees Celsius.

Choice A is incorrect. This statement represents a situation where x , not y , represents the boiling point, in degrees Celsius, and y , not x , represents the molecular weight, in grams, per mole.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question Difficulty:

Question ID 0a74365c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 0a74365c

The population of the town of Smithville doubled every **75** years from **1659** to **1959**. The population of this town was **240,000** in **1959**. What was the population of this town in **1659**?

ID: 0a74365c Answer

Correct Answer:
15000

Rationale

The correct answer is **15,000**. Let **y** represent the population of Smithville **x** years after **1659**. Since the population doubled every **75** years, the relationship between **x** and **y** can be represented by an exponential equation in the form $y = a(b)^{\frac{x}{k}}$, where **a** is the population of Smithville in **1659** and the population increases by a factor of **b** every **k** years. It's given that the population doubled, or increased by a factor of **2**, every **75** years. Therefore, **b** = **2** and **k** = **75**. Substituting **2** for **b** and **75** for **k** in the equation $y = a(b)^{\frac{x}{k}}$ yields $y = a(2)^{\frac{x}{75}}$. It's also given that the population of this town was **240,000** in **1959**. The number of years from **1659** to **1959** is **1959** – **1659**, or **300** years. Therefore, when **x** = **300**, the value of **y** is **240,000**. Substituting **300** for **x** and **240,000** for **y** in the equation $y = a(2)^{\frac{x}{75}}$ yields $240,000 = a(2)^{\frac{300}{75}}$, or $240,000 = a(2)^4$, which is equivalent to $240,000 = a(16)$. Dividing both sides of this equation by **16** yields **15,000** = **a**. Therefore, the population of this town in **1659** was **15,000**.

Question Difficulty:
Medium

Question ID 565428e1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: 565428e1

$$\sqrt[3]{p^2} = t^{\frac{9}{7}}$$

In the given equation, $p > 1$ and $t > 1$. If $t = p^{3n-1}$, where n is a constant, what is the value of n ?

ID: 565428e1 Answer

Correct Answer:
.5061, .5062, 41/81

Rationale

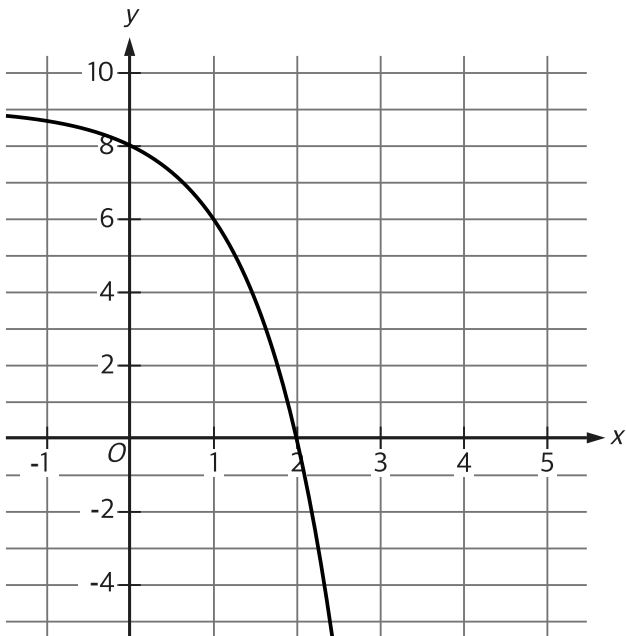
The correct answer is $\frac{41}{81}$. An expression of the form $\sqrt[n]{a^m}$, where m and n are integers greater than 1 and $a \geq 0$, is equivalent to $a^{\frac{m}{n}}$. Therefore, the expression on the left-hand side of the given equation is equivalent to $p^{\frac{2}{3}}$; thus, $p^{\frac{2}{3}} = t^{\frac{9}{7}}$. If $t = p^{3n-1}$, where n is a constant, then substituting p^{3n-1} for t in the equation yields $p^{\frac{2}{3}} = p^{\frac{9}{7}(3n-1)}$. It's given that $p > 1$; therefore, $\frac{2}{3} = \frac{9}{7}(3n - 1)$. Multiplying each side of this equation by 21 yields $14 = 27(3n - 1)$. Distributing multiplication over subtraction yields $14 = 81n - 27$. Adding 27 to each side of this equation yields $41 = 81n$. Dividing each side of this equation by 81 yields $\frac{41}{81} = n$. Therefore, if $t = p^{3n-1}$, where n is a constant, the value of n is $\frac{41}{81}$. Note that 41/81, .5061, .5062, and 0.506 are examples of ways to enter a correct answer.

Question Difficulty:
Hard

Question ID 36b6f8ba

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 36b6f8ba



The graph of $y = f(x) + 4$ is shown. Which equation defines function f ?

- A. $f(x) = -3^x + 1$
- B. $f(x) = -3^x + 5$
- C. $f(x) = -3^x + 8$
- D. $f(x) = -3^x + 9$

ID: 36b6f8ba Answer

Correct Answer:
B

Rationale

Choice B is correct. The graph of $y = f(x) + 4$ is shown, and the graph passes through the point $(0, 8)$. Substituting 0 for x and 8 for y in this equation yields $8 = f(0) + 4$. Subtracting 4 from both sides of this equation yields $4 = f(0)$. Therefore, when $x = 0$, the value of $f(x)$ is 4 . Each of the given choices is in the form $f(x) = -3^x + k$, where k is a constant. Substituting 0 for x and 4 for $f(x)$ in this equation yields $4 = -3^0 + k$, or $4 = -1 + k$. Adding 1 to both sides of this equation yields $5 = k$. Substituting 5 for k in the equation $f(x) = -3^x + k$ yields $f(x) = -3^x + 5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID 16aa9758

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 16aa9758

$$f(x) = -6x^2 + 60x - 126$$

The function f is defined by the given equation. Which of the following equivalent forms of the equation displays the maximum value of the function as a constant or coefficient?

- A. $f(x) = -6x^2 + 42x + 18x - 126$
- B. $f(x) = -6x(x - 7) + 18(x - 7)$
- C. $f(x) = -6(x - 5)^2 + 24$
- D. $f(x) = -6(x - 7)(x - 3)$

ID: 16aa9758 Answer

Correct Answer:

C

Rationale

Choice C is correct. The given equation is quadratic. The maximum value of a function defined by a quadratic equation can be displayed as a constant in the vertex form of the equation, $f(x) = a(x - h)^2 + k$, where the maximum value of the function is k , which occurs when $x = h$, and a is a constant. The given equation can be rewritten in this form by completing the square. To complete the square, the given equation can be rewritten as $f(x) = -6(x^2 - 10x) - 126$, which is equivalent to $f(x) = -6(x^2 - 10x + 25) - 126 + 6(25)$, or $f(x) = -6(x - 5)^2 + 24$. This equation is in vertex form, where $a = -6$, $h = 5$, and $k = 24$. Therefore, $f(x) = -6(x - 5)^2 + 24$ displays the maximum value, 24 , of the function as a constant.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This form displays one of the zeros, not the maximum value, of the function as a constant.

Choice D is incorrect. This form displays the zeros, not the maximum value, of the function as constants.

Question Difficulty:

Hard

Question ID 2f8a580c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 2f8a580c

The function k is defined by $k(x) = x^3 + 110$. What is the value of $k(x)$ when $x = 2$?

- A. 118
- B. 116
- C. 115
- D. 110

ID: 2f8a580c Answer

Correct Answer:
A

Rationale

Choice A is correct. Substituting 2 for x in the given equation yields $k(2) = 2^3 + 110$, which is equivalent to $k(2) = 8 + 110$, or $k(2) = 118$. Therefore, the value of $k(x)$ when $x = 2$ is 118.

Choice B is incorrect. This is the value of $k(x)$ when $x = 2$ if $k(x) = 3x + 110$, not $k(x) = x^3 + 110$.

Choice C is incorrect. This is the value of $k(x)$ when $x = 2$ if $k(x) = x + 3 + 110$, not $k(x) = x^3 + 110$.

Choice D is incorrect. This is the value of $k(x)$ when $x = 0$, not $x = 2$.

Question Difficulty:
Easy

Question ID b3b07f0c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: b3b07f0c

x	24	30	32
$f(x)$	-8	-8	8

For the quadratic function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f ?

- A. $f(x) = (x - 24)(x - 30) + 4$
- B. $f(x) = (x - 24)(x - 30) - 8$
- C. $f(x) = (x - 8)(x - 32) + 32$
- D. $f(x) = (x - 8)(x - 32) - 32$

ID: b3b07f0c Answer

Correct Answer:

B

Rationale

Choice B is correct. According to the table, $f(24) = -8$, $f(30) = -8$, and $f(32) = 8$. Let the quadratic function g be defined by $g(x) = f(x) + 8$. Therefore, $g(24) = f(24) + 8$, $g(30) = f(30) + 8$, and $g(32) = f(32) + 8$. Substituting -8 for $f(24)$ in the equation $g(24) = f(24) + 8$ yields $g(24) = -8 + 8$, or $g(24) = 0$. Substituting -8 for $f(30)$ in the equation $g(30) = f(30) + 8$ yields $g(30) = -8 + 8$, or $g(30) = 0$. Substituting 8 for $f(32)$ in the equation $g(32) = f(32) + 8$ yields $g(32) = 8 + 8$, or $g(32) = 16$. By the factor theorem, if $g(24) = 0$ and $g(30) = 0$, then $g(x) = k(x - 24)(x - 30)$, where k is a constant. Since $g(32) = 16$, substituting 32 for x and 16 for $g(x)$ in the equation $g(x) = k(x - 24)(x - 30)$ yields $16 = k(32 - 24)(32 - 30)$, or $16 = k(8)(2)$, which is equivalent to $16 = 16k$. Dividing both sides of this equation by 16 yields $1 = k$. Substituting 1 for k in the equation $g(x) = k(x - 24)(x - 30)$ yields $g(x) = (x - 24)(x - 30)$. Since $g(x) = f(x) + 8$, it follows that $g(x) - 8 = f(x)$. Subtracting 8 from both sides of the equation $g(x) = (x - 24)(x - 30)$ yields $g(x) - 8 = (x - 24)(x - 30) - 8$. Substituting $f(x)$ for $g(x) - 8$ in this equation yields $f(x) = (x - 24)(x - 30) - 8$.

Choice A is incorrect. If this equation defined f , then when $x = 24$, the value of $f(x)$ would be 4 , not -8 .

Choice C is incorrect. If this equation defined f , then when $x = 32$, the value of $f(x)$ would be 32 , not 8 .

Choice D is incorrect. If this equation defined f , then when $x = 32$, the value of $f(x)$ would be -32 , not 8 .

Question Difficulty:

Medium

Question ID 63c00e2b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 63c00e2b

$\frac{r}{38} = s + t$

The given equation relates the quantities r , s , and t . Which equation correctly expresses r in terms of s and t ?

- A. $r = s + t + 38$
- B. $r = s + t - 38$
- C. $r = 38(s + t)$
- D. $r = \frac{s+t}{38}$

ID: 63c00e2b Answer

Correct Answer:

C

Rationale

Choice C is correct. Multiplying both sides of the given equation by 38 yields $r = 38(s + t)$.

Choice A is incorrect. This equation is equivalent to $r - 38 = s + t$, not $\frac{r}{38} = s + t$.

Choice B is incorrect. This equation is equivalent to $r + 38 = s + t$, not $\frac{r}{38} = s + t$.

Choice D is incorrect. This equation is equivalent to $38r = s + t$, not $\frac{r}{38} = s + t$.

Question Difficulty:

Easy

Question ID 0f9f8ea7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 0f9f8ea7

$$y = 3,600(a)^x$$

The given equation, where a is a positive constant, gives the predicted number of bacteria, y , in a growth medium x hours after the number of bacteria was initially measured. According to the equation, what was the predicted number of bacteria initially measured in the growth medium?

ID: 0f9f8ea7 Answer

Correct Answer:
3600

Rationale

The correct answer is **3,600**. The predicted number of bacteria initially measured in the growth medium is the value of y when $x = 0$. Substituting **0** for x in the given equation yields $y = 3,600(a)^0$. Since a is positive, this equation is equivalent to $y = 3,600(1)$. Therefore, when $x = 0$, the value of y is **3,600**. Thus, the predicted number of bacteria initially measured in the growth medium is **3,600**.

Question Difficulty:
Medium

Question ID c8a5c221

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: c8a5c221

$$(9x^4 + 8x^3) - (7x^3 + 5x^2) - (3x^2 + x)$$

The given expression can be written in the form $9x^4 + x^3 + ax^2 - x$, where a is a constant. What is the value of a ?

- A. -8
- B. -2
- C. 2
- D. 8

ID: c8a5c221 Answer

Correct Answer:
A

Rationale

Choice A is correct. The given expression is equivalent to $9x^4 + 8x^3 - 7x^3 - 5x^2 - 3x^2 - x$. By combining like terms, this can be written as $9x^4 + x^3 - 8x^2 - x$. This expression is in the form $9x^4 + x^3 + ax^2 - x$, where a is a constant. Therefore, the value of a is -8 .

Choice B is incorrect. This may result from adding $-5x^2$ and $3x^2$ rather than $-5x^2$ and $-3x^2$.

Choice C is incorrect. This may result from adding $5x^2$ and $-3x^2$ rather than $-5x^2$ and $-3x^2$.

Choice D is incorrect. This may result from adding $5x^2$ and $3x^2$ rather than $-5x^2$ and $-3x^2$.

Question Difficulty:
Hard

Question ID 041eda89

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 041eda89

$x^2 + 7x + 5 = 0$

One solution to the given equation can be written as $x = \frac{-7+\sqrt{k}}{2}$, where k is a constant. What is the value of k ?

ID: 041eda89 Answer

Correct Answer:
29

Rationale

The correct answer is **29**. The solutions to a quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be calculated using the quadratic formula, which gives two solutions: $x = \frac{-b+\sqrt{b^2-4ac}}{2a}$ and $x = \frac{-b-\sqrt{b^2-4ac}}{2a}$. The given equation is in the form $ax^2 + bx + c = 0$, where $a = 1$, $b = 7$, and $c = 5$. It follows that one solution to the given quadratic equation is $x = \frac{-7+\sqrt{7^2-4(1)(5)}}{2(1)}$, or $x = \frac{-7+\sqrt{49-20}}{2}$, which is equivalent to $x = \frac{-7+\sqrt{29}}{2}$. It's given that one solution to the given equation can be written as $x = \frac{-7+\sqrt{k}}{2}$, where k is a constant. It follows that the value of k is **29**.

Question Difficulty:
Hard

Question ID 4d425cef

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 4d425cef

For the polynomial function f , the graph of $y = f(x)$ in the xy -plane passes through the points $(-5, 0)$, $(1, 0)$, and $(4, 0)$. Which of the following must be a factor of $f(x)$?

- A. $x + 1$
- B. $x + 4$
- C. $x - 1$
- D. $x - 5$

ID: 4d425cef Answer

Correct Answer:

C

Rationale

Choice C is correct. It's given that f is a polynomial function and the graph of $y = f(x)$ in the xy -plane passes through the points $(-5, 0)$, $(1, 0)$, and $(4, 0)$. A point in the form $(a, 0)$ on the graph of $f(x)$ means that $x - a$ is a factor of $f(x)$. Since the graph of the function f passes through the point $(1, 0)$, $x - 1$ must be a factor of $f(x)$.

Choice A is incorrect. This is a factor if the graph of $y = f(x)$ passed through $(-1, 0)$.

Choice B is incorrect. This is a factor if the graph of $y = f(x)$ passed through $(-4, 0)$.

Choice D is incorrect. This is a factor if the graph of $y = f(x)$ passed through $(5, 0)$.

Question Difficulty:

Medium

Question ID 3934dd0a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: 3934dd0a

Which quadratic equation has exactly one distinct real solution?

- A. $(x + 15)^2 = 0$
- B. $(x + 15)^2 = -45$
- C. $(x + 15)^2 = 45$
- D. $(x + 15)^2 = 135$

ID: 3934dd0a Answer

Correct Answer:
A

Rationale

Choice A is correct. Taking the square root of both sides of $(x + 15)^2 = 0$ yields $x + 15 = 0$. Subtracting 15 from both sides of this equation yields $x = -15$. Therefore, this equation has exactly one distinct real solution.

Choice B is incorrect. This equation has no distinct real solutions, not exactly one distinct real solution.

Choice C is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution.

Choice D is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution.

Question Difficulty:
Medium

Question ID dc67c130

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	

ID: dc67c130

Which expression is equivalent to $64t^2s^3 - 56t^3s$?

- A. $4ts(16s^2 - 14ts)$
- B. $4ts(16t^2s^2 - 14t)$
- C. $4t^2s(16ts - 14s)$
- D. $4t^2s(16s^2 - 14t)$

ID: dc67c130 Answer

Correct Answer:
D

Rationale

Choice D is correct. For the expression $64t^2s^3 - 56t^3s$, 4 is a common factor of 64 and 56 , t^2 is a common factor of t^2 and t^3 , and s is a common factor of s^3 and s . It follows that $4t^2s$ is a common factor of each term in the given expression. Factoring out $4t^2s$ from each term in the expression $64t^2s^3 - 56t^3s$ yields $4t^2s(16s^2 - 14t)$. Therefore, $4t^2s(16s^2 - 14t)$ is equivalent to $64t^2s^3 - 56t^3s$.

Choice A is incorrect. This expression is equivalent to $64ts^3 - 56t^2s^2$, not $64t^2s^3 - 56t^3s$.

Choice B is incorrect. This expression is equivalent to $64t^3s^3 - 56t^2s$, not $64t^2s^3 - 56t^3s$.

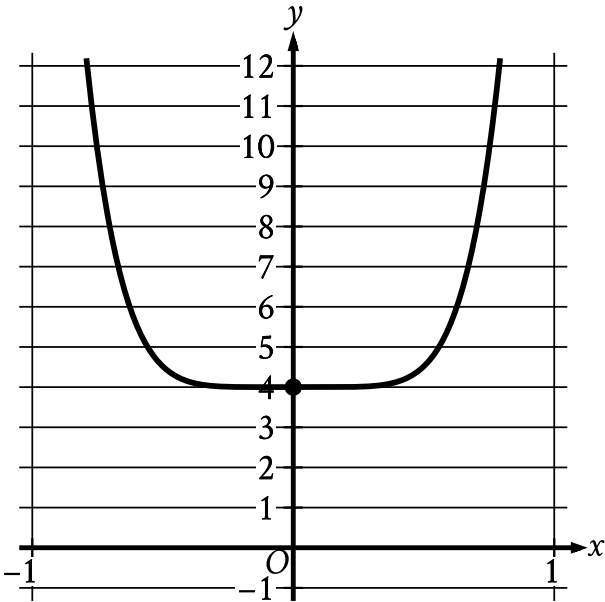
Choice C is incorrect. This expression is equivalent to $64t^3s^2 - 56t^2s^2$, not $64t^2s^3 - 56t^3s$.

Question Difficulty:
Easy

Question ID 5f07b257

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 5f07b257



The graph of the polynomial function f , where $y = f(x)$, is shown. The y -intercept of the graph is $(0, y)$. What is the value of y ?

ID: 5f07b257 Answer

Correct Answer:

4

Rationale

The correct answer is 4. The y -intercept of the graph of a function in the xy -plane is the point where the graph intersects the y -axis. The graph of the polynomial function shown intersects the y -axis at the point $(0, 4)$. It's given that the y -intercept of the graph is $(0, y)$. Therefore, the value of y is 4.

Question Difficulty:

Easy

Question ID fc2ec559

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: fc2ec559

$$\begin{aligned}x + 5 &= 14 \\ y &= 4x^2 + 4\end{aligned}$$

At what point (x, y) do the graphs of the equations in the given system intersect?

- A. $(9, 324)$
- B. $(9, 328)$
- C. $(14, 4)$
- D. $(14, 788)$

ID: fc2ec559 Answer

Correct Answer:
B

Rationale

Choice B is correct. The point (x, y) at which the graphs of the given equations intersect is the solution to the system of equations. Subtracting 5 from both sides of the equation $x + 5 = 14$ yields $x = 9$. Substituting 9 for x in the equation $y = 4x^2 + 4$ yields $y = 4(9)^2 + 4$, or $y = 328$. Therefore, the graphs of the equations in the given system intersect at the point $(9, 328)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID faeae876

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: faeae876

$52(x^3 + 64)(x^4 - 81) = 0$

How many distinct real solutions does the given equation have?

- A. Exactly two
- B. Exactly three
- C. Exactly five
- D. Exactly seven

ID: faeae876 Answer

Correct Answer:

B

Rationale

Choice B is correct. The given equation can be rewritten as $52(x + 4)(x^2 - 4x + 16)(x^2 - 9)(x^2 + 9) = 0$, which is equivalent to $52(x + 4)(x^2 - 4x + 16)(x - 3)(x + 3)(x^2 + 9) = 0$. Applying the zero product property in this equation yields $52 = 0$, $x + 4 = 0$, $x^2 - 4x + 16 = 0$, $x - 3 = 0$, $x + 3 = 0$, and $x^2 + 9 = 0$. The equation $52 = 0$ has no solution. Subtracting 4 from both sides of the equation $x + 4 = 0$ yields $x = -4$. Adding 3 to both sides of the equation $x - 3 = 0$ yields $x = 3$. Subtracting 3 from both sides of the equation $x + 3 = 0$ yields $x = -3$. The discriminant of a quadratic equation of the form $ax^2 + bx + c = 0$ is $b^2 - 4ac$. If the discriminant is positive, the equation has two distinct real solutions. If the discriminant is zero, the equation has one distinct real solution. If the discriminant is negative, the equation has no real solutions. In the equation $x^2 - 4x + 16 = 0$, $a = 1$, $b = -4$, and $c = 16$. Substituting these values in the equation for the discriminant, $b^2 - 4ac$, yields $(-4)^2 - 4(1)(16)$, or -48 . Since the discriminant is negative, this equation has no real solutions. In the equation $x^2 + 9 = 0$, $a = 1$, $b = 0$, and $c = 9$. Substituting these values in the equation for the discriminant, $b^2 - 4ac$, yields $0^2 - 4(1)(9)$, or -36 . Since the discriminant is negative, this equation has no real solutions. Therefore, the distinct real solutions to the given equation are $x = -4$, $x = 3$, and $x = -3$. Thus, the given equation has exactly three distinct real solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Medium

Question ID cec36caf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	

ID: cec36caf

$8x^2 - 40 = 32$

What is the positive solution to the given equation?

- A. 3
- B. 4
- C. 9
- D. 72

ID: cec36caf Answer

Correct Answer:

A

Rationale

Choice A is correct. Adding 40 to both sides of the given equation yields $8x^2 = 72$. Dividing both sides of this equation by 8 yields $x^2 = 9$. Taking the square root of both sides of this equation yields $x = 3$ or $x = -3$. Therefore, the positive solution to the given equation is 3.

Choice B is incorrect. This is the solution to the equation $8x = 32$, not the given equation.

Choice C is incorrect. This is the solution to the equation $8x - 40 = 32$, not the given equation.

Choice D is incorrect. This is the solution to the equation $x - 40 = 32$, not the given equation.

Question Difficulty:

Easy

Question ID 3e28a622

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: 3e28a622

The function g is defined by $g(x) = \sqrt{x} + 300$. What is the value of $g(x)$ when $x = 81$?

- A. 9
- B. 300
- C. 309
- D. 381

ID: 3e28a622 Answer

Correct Answer:
C

Rationale

Choice C is correct. It's given that function g is defined by $g(x) = \sqrt{x} + 300$. Substituting 81 for x in function g yields $g(81) = \sqrt{81} + 300$, which is equivalent to $g(81) = 9 + 300$, or $g(81) = 309$. Therefore, the value of $g(x)$ when $x = 81$ is 309 .

Choice A is incorrect. This is the value of $\sqrt{81}$, not $\sqrt{81} + 300$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $81 + 300$, not $\sqrt{81} + 300$.

Question Difficulty:
Easy

Question ID 555939d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: 555939d2

A cable provider wanted to know how many of its **30,000** customers would be interested in a new service plan. The provider selected **300** customers at random and asked each customer whether the customer would be interested in the new plan. Of those surveyed, **8** said they would be interested. Which of the following is the best estimate of the total number of customers who would be interested in the new service plan?

- A. 8
- B. 80
- C. 800
- D. 8,000

ID: 555939d2 Answer

Correct Answer:
C

Rationale

Choice C is correct. It’s given that **8** out of **300** customers in a random sample said they would be interested in the new plan. This means the fraction of sampled customers who said they would be interested is $\frac{8}{300}$. Multiplying this fraction by the total customer population yields $(\frac{8}{300})(30,000)$, or **800**. Therefore, the best estimate of the total number of customers who would be interested in the new service plan is **800**.

Choice A is incorrect. This is the number of customers surveyed who said they would be interested in the new plan, not the best estimate of the total number of customers who would be interested in the new plan.

Choice B is incorrect and may result from conceptual or calculation errors.

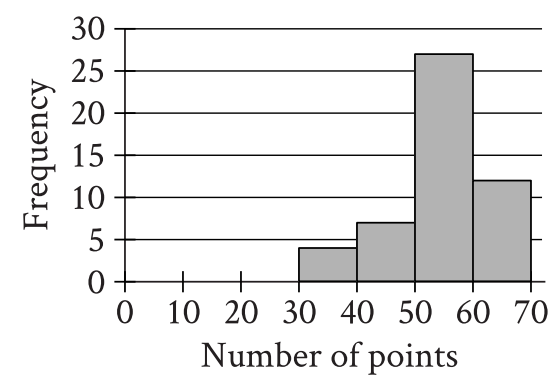
Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy

Question ID 608cf8e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 608cf8e2



The histogram summarizes data set A, which represents the number of points per player earned by **50** players of a game. A new player earns **18** points playing the game, and this number of points is added to data set A to create data set B with **51** values. Which of the following must be true?

- I. The median number of points per player for data set B is less than the median number of points per player for data set A.

II. The mean number of points per player for data set B is less than the mean number of points per player for data set A.
- A. I only

B. II only

C. I and II

D. Neither I nor II

ID: 608cf8e2 Answer

Correct Answer:
B

Rationale

Choice B is correct. The mean of a data set is the sum of the values divided by the number of values. It's given that a new player earns **18** points playing the game, and this number of points is added to data set A to create data set B with **51** values. The histogram shows that all the values in data set A are greater than **18**. Since the additional number of points, **18**, is less than any of the values in the original data set, the mean number of points per player for data set B is less than the mean number of points per player for data set A. The median of a data set with **50** values is between the **25th** and **26th** values when the values are listed in ascending order. The median of a data set with **51** values is the **26th** value when the values are listed in ascending order. Therefore, the median of data set B is the **26th** value in data set B, which is the **25th** value in data set A. The histogram shows that the **25th** and **26th** values in data set A are both between **50** and **60** points. Therefore, the **25th** and **26th** values in data set A could be equal, in which case the median of data set A would be equal to both the **25th** and **26th** values in data set A. Then, the median number of points per player for data set A would be equal to the median number of points per player for data set B. Therefore, only statement II must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

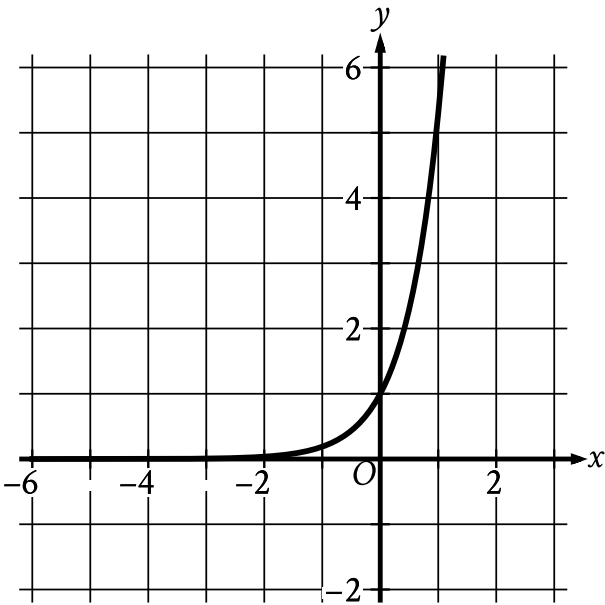
Hard

Question ID 632cf8f9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 632cf8f9

The graph of the function f is shown, where $y = f(x)$.



Which of the following best describes the function f ?

- A. Decreasing exponential
- B. Increasing exponential
- C. Decreasing linear
- D. Increasing linear

ID: 632cf8f9 Answer

Correct Answer:
B

Rationale

Choice B is correct. The graph of function f shows that as x increases, $f(x)$ also increases, which means that f is an increasing function. The graph of function f is curved upward and increases at an increasing rate, not a constant rate. This behavior is characteristic of exponential functions, not linear functions. Therefore, of the given choices, function f is best described as increasing exponential.

Choice A is incorrect. For a decreasing function, as x increases, $f(x)$ decreases rather than increases.

Choice C is incorrect. For a decreasing function, as x increases, $f(x)$ decreases rather than increases, and the graph of a linear function isn't curved.

Choice D is incorrect. The graph of a linear function isn't curved.

Question Difficulty:

Question ID b5bc47d5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: b5bc47d5

In a scale model of an office building, a length of **1** inch is equivalent to a length of $\frac{17}{3}$ feet in the actual office building. If a length in the actual office building measures x feet, which expression represents the corresponding length, in inches, in the model in terms of x ?

- A. $\frac{3x}{(17)(12)}$
- B. $\frac{17x}{(12)(3)}$
- C. $\frac{3x}{17}$
- D. $\frac{17x}{3}$

ID: b5bc47d5 Answer

Correct Answer:

C

Rationale

Choice C is correct. It's given that in a scale model, a length of **1** inch is equivalent to a length of $\frac{17}{3}$ feet in the actual office building. This means that the ratio of a length, in feet, in the actual office building to the corresponding length, in inches, in the model is $\frac{17}{3}$ to **1**. Therefore, the length, in inches, in the model corresponding to a length in the actual office building of x feet can be found by dividing x by $\frac{17}{3}$. Dividing by $\frac{17}{3}$ is equivalent to multiplying by $\frac{3}{17}$, so the corresponding length, in inches, in the model in terms of x can be represented by the expression $\frac{3x}{17}$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. This expression represents the corresponding length, in inches, in the model in terms of x for a model where **1** inch in the model is equivalent to a length of $\frac{3}{17}$, not $\frac{17}{3}$, feet in the actual building.

Question Difficulty:

Hard

Question ID 699af7b3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 699af7b3

The weight of sample A is **468%** of the weight of sample B, and the weight of sample A is **0.780%** of the weight of sample C. If the weight of sample C is ***p*%** of the weight of sample B, what is the value of ***p***?

ID: 699af7b3 Answer

Correct Answer:
60000

Rationale

The correct answer is **60,000**. It's given that the weight of sample A is **468%** of the weight of sample B and **0.780%** of the weight of sample C. Let the weight of sample B be ***b*** and the weight of sample C be ***c***. Then, the weight of sample A is equivalent to $\frac{468b}{100}$, or **4.68*b***, and also equivalent to $\frac{0.780c}{100}$, or **0.00780*c***. Setting these expressions equal to each other yields **4.68*b* = 0.00780*c***. Dividing both sides of this equation by **0.00780** yields $\frac{4.68b}{0.00780} = c$, or ***c* = 600*b***. If the weight of sample C is ***p*%** of the weight of sample B, then ***c* = $\frac{p}{100}b$** . Since ***c* = 600*b*** and ***c* = $\frac{p}{100}b$** , it follows that **600*b* = $\frac{p}{100}b$** , or **600 = $\frac{p}{100}$** . Multiplying both sides of this equation by **100** yields **60,000 = *p***. Therefore, the value of ***p*** is **60,000**.

Question Difficulty:
Hard

Question ID 40f0ee70

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 40f0ee70

Task	Time (minutes)
I	5
II	2
III	15
IV	9
V	9

The table shows the amount of time it took Robert to complete each of the 5 tasks in a project. What was the mean time, in minutes, for Robert to complete a task for this project?

ID: 40f0ee70 Answer

Correct Answer:
8

Rationale

The correct answer is 8. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. It's given in the table that the times, in minutes, it took Robert to complete each of the tasks are 5, 2, 15, 9, and 9. The sum of these times is $5 + 2 + 15 + 9 + 9$, or 40, minutes. There are 5 tasks in the project. Therefore, the mean time, in minutes, for Robert to complete a task for this project is $40 \div 5$, or 8.

Question Difficulty:
Easy

Question ID 58ba7239

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 58ba7239

The number a is **55%** less than the number b . The number b is **320%** greater than **160**. What is the value of a ?

ID: 58ba7239 Answer

Correct Answer:
1512/5, 302.4

Rationale

The correct answer is **302.4**. It's given that the number b is **320%** greater than **160**. It follows that b is equal to **160** plus **320%** of **160**, which can be written as $b = 160 + 3.2(160)$, or $b = 672$. It's also given that the number a is **55%** less than the number b . It follows that a is equal to b minus **55%** of b , which can be written as $a = b - 0.55b$, or $a = 0.45b$. Substituting **672** for b in this equation yields $a = 0.45(672)$, or $a = 302.4$.

Question Difficulty:
Hard

Question ID 9ce8cbbe

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: 9ce8cbbe

Set K consists of all the positive integers that are less than or equal to **160**, where none of the integers are repeated. If a number is selected at random from set K, what is the probability of selecting a number that is even and less than or equal to **50**?

- A. $\frac{5}{32}$
- B. $\frac{3}{16}$
- C. $\frac{5}{21}$
- D. $\frac{5}{16}$

ID: 9ce8cbbe Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that set K consists of all the positive integers that are less than or equal to **160**, where none of the integers are repeated. Therefore, set K consists of the integers **1** through **160**, which is **160** numbers. Of these, the integers **1** through **50** are less than or equal to **50**. Therefore, there are **50** numbers that belong to set K that are less than or equal to **50**. Of these, half are even. Therefore, $\frac{50}{2}$, or **25**, numbers that belong to set K are even and less than or equal to **50**. The probability of selecting at random a number from set K that is even and less than or equal to **50** is equal to the number of numbers that belong to set K that are even and less than or equal to **50**, which is **25**, divided by the total number of numbers that belong to set K, which is **160**. It follows that the probability of selecting at random from set K a number that is even and less than or equal to **50** is $\frac{25}{160}$, or $\frac{5}{32}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the probability of selecting at random a number from set K that is less than or equal to **50**, not a number that is even and less than or equal to **50**.

Question Difficulty:

Hard

Question ID c74df824

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	

ID: c74df824

An area of **46.00** square nautical miles is equivalent to ***k*** square kilometers. To the nearest tenth, what is the value of ***k***?
(**1 nautical mile = 1.852 kilometers**)

ID: c74df824 Answer

Correct Answer:
157.8, 789/5

Rationale

The correct answer is **157.8**. It's given that an area of **46.00** square nautical miles is equivalent to ***k*** square kilometers. Since **1** nautical mile is equal to **1.852** kilometers, it follows that an area of **1** square nautical mile is equivalent to **$(1.852)^2$** , or **3.429904**, square kilometers. Multiplying **46.00** by **3.429904** yields **157.775584** square kilometers. Rounding this value to the nearest tenth yields **157.8**. Therefore, the value of ***k*** is **157.8**.

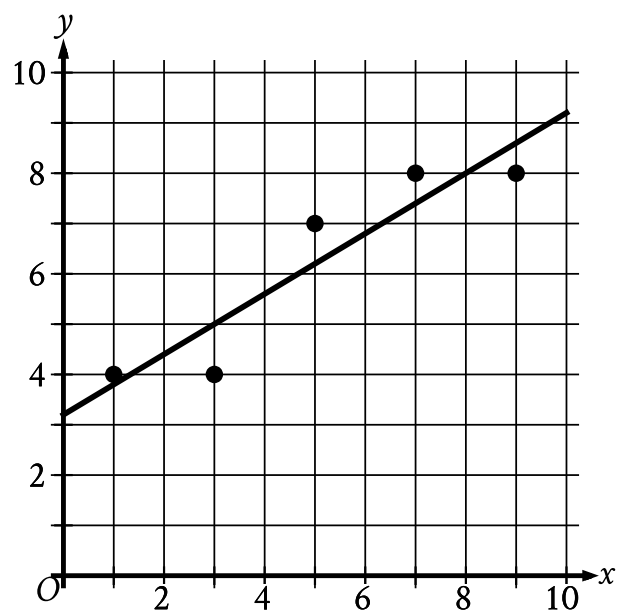
Question Difficulty:
Hard

Question ID 38525983

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: 38525983

The scatterplot shows the relationship between x and y . A line of best fit is also shown.



Which of the following is closest to the slope of this line of best fit?

- A. 0.60
- B. 2.50
- C. 7.80
- D. 8.00

ID: 38525983 Answer

Correct Answer:
A

Rationale

Choice A is correct. A line in the xy -plane that passes through the points (x_1, y_1) and (x_2, y_2) has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes through the points with approximate coordinates $(0, 3.2)$ and $(2, 4.4)$. It follows that the slope of the line of best fit is approximately $\frac{4.4 - 3.2}{2 - 0}$, which is equivalent to $\frac{1.2}{2}$, or 0.6. Therefore, of the given choices, 0.6 is closest to the slope of the line of best fit shown.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Question ID 73362607

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	

ID: 73362607

Amara grows cherry tomatoes in her backyard. This year, she harvested **750** cherry tomatoes and gave **10%** of them to her neighbor. How many of the harvested cherry tomatoes did Amara give to her neighbor?

ID: 73362607 Answer

Correct Answer:

75

Rationale

The correct answer is **75**. It’s given that Amara harvested **750** cherry tomatoes and gave **10%** of them to her neighbor. **10%** can be written as $\frac{10}{100}$, or **0.1**. Therefore, the number of cherry tomatoes that Amara gave to her neighbor can be calculated by multiplying the total number of cherry tomatoes by **0.1**, which gives **(0.1)(750)**, or **75**.

Question Difficulty:

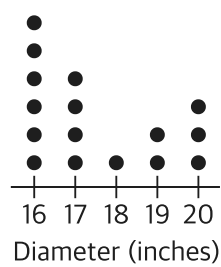
Easy

Question ID 7b1a3200

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: 7b1a3200

To study the characteristics of sea stars in a group of tide pools, researchers measured the diameter of the sea stars within the tide pools. The dot plot gives the diameter, to the nearest inch, of each of the sea stars in these tide pools.



Based on the dot plot, how many sea stars had a diameter, to the nearest inch, of **16** inches?

- A. **16**
- B. **6**
- C. **4**
- D. **1**

ID: 7b1a3200 Answer

Correct Answer:
B

Rationale

Choice B is correct. The dot plot gives the diameter, to the nearest inch, of each sea star in a group of tide pools. The number of dots above a given diameter represents the number of sea stars with that diameter, to the nearest inch. There are **6** dots above **16** inches. Therefore, **6** sea stars had a diameter, to the nearest inch, of **16** inches.

Choice A is incorrect and may result from calculation or conceptual errors.

Choice C is incorrect. This is the number of sea stars with a diameter, to the nearest inch, of **17** inches, not **16** inches.

Choice D is incorrect. This is the number of sea stars with a diameter, to the nearest inch, of **18** inches, not **16** inches.

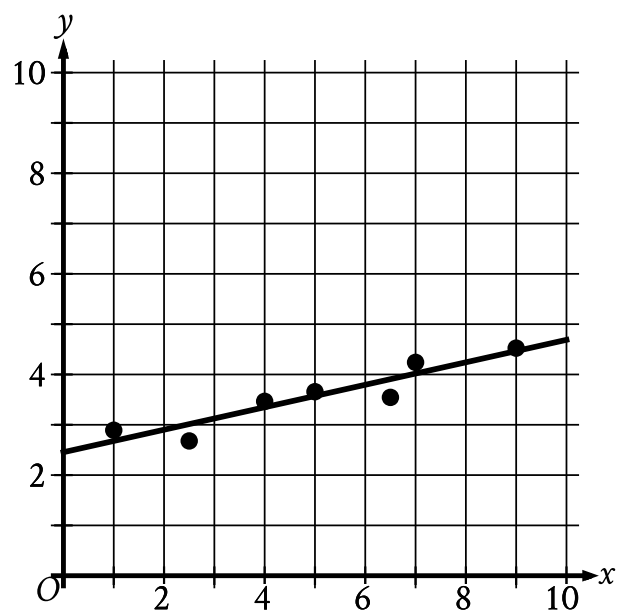
Question Difficulty:
Easy

Question ID e9129a5c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	

ID: e9129a5c

In the given scatterplot, a line of best fit for the data is shown.



Which of the following is closest to the slope of the line of best fit?

- A. -2.45
- B. -0.22
- C. 0.22
- D. 2.45

ID: e9129a5c Answer

Correct Answer:
C

Rationale

Choice C is correct. A line in the xy -plane that passes through the points (x_1, y_1) and (x_2, y_2) has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes through the points with approximate coordinates $(0, 2.5)$ and $(10, 4.7)$. It follows that the slope of the line of best fit is approximately $\frac{4.7 - 2.5}{10 - 0}$, which is equivalent to $\frac{2.2}{10}$, or 0.22 . Therefore, of the given choices, **0.22** is closest to the slope of the line of best fit.

Choice A is incorrect. This is closest to the negative y -coordinate of the y -intercept, not to the slope, of the line of best fit.

Choice B is incorrect. This is closest to the negative slope, not to the slope, of the line of best fit.

Choice D is incorrect. This is closest to the y -coordinate of the y -intercept, not to the slope, of the line of best fit.

Question Difficulty:

Question ID c78bb2b8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	

ID: c78bb2b8

A scientist measured the lengths of **240** gray seals from Muskeget Island and **120** gray seals from Sable Island. The scientist determined that the mean length of the **240** gray seals from Muskeget Island was **88** inches and the mean length of the **120** gray seals from Sable Island was **94** inches. What was the mean length of all **360** gray seals the scientist measured for this study?

- A. **89**
- B. **90**
- C. **91**
- D. **92**

ID: c78bb2b8 Answer

Correct Answer:
B

Rationale

Choice B is correct. The mean of a data set is calculated by dividing the sum of the values in the data set by the number of values in the data set. It’s given that the mean length of the **240** gray seals from Muskeget Island was **88** inches. This can be represented by the equation $\frac{x}{240} = 88$, where x represents the sum of the lengths, in inches, of the **240** gray seals from Muskeget Island. Multiplying both sides of this equation by **240** yields $x = (240)(88)$, or $x = 21,120$ inches. It’s also given that the mean length of the **120** gray seals from Sable Island was **94** inches. This can be represented by the equation $\frac{y}{120} = 94$, where y represents the sum of the lengths, in inches, of the **120** gray seals from Sable Island. Multiplying both sides of this equation by **120** yields $y = (120)(94)$, or $y = 11,280$ inches. Therefore, the sum of the lengths, in inches, of all **360** gray seals is **21,120 + 11,280**, or **32,400**. Dividing this sum by **360** yields $\frac{32,400}{360}$, or **90** inches. Therefore, the mean length of all **360** gray seals the scientist measured for this study is **90** inches.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the mean length of **88** inches and **94** inches, not the mean length of all **360** gray seals the scientist measured for this study.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID ecf8d15f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	

ID: ecf8d15f

A total of **50** children attended a summer camp and were offered **4** types of sandwiches. The table shows the number of children who chose each type of sandwich.

Type of sandwich	Number of children
Turkey	15
Chicken	23
Ham	3
Vegetarian	9
Total	50

If one of these children is selected at random, what is the probability of selecting a child who chose a vegetarian sandwich?

- A. $\frac{9}{100}$
- B. $\frac{9}{50}$
- C. $\frac{1}{4}$
- D. $\frac{9}{10}$

ID: ecf8d15f Answer

Correct Answer:

B

Rationale

Choice B is correct. If one of these children is selected at random, the probability of selecting a child who chose a vegetarian sandwich is equal to the number of children who chose a vegetarian sandwich divided by the total number of children. According to the table, there are a total of **50** children, and **9** of these children chose a vegetarian sandwich. Therefore, the probability of selecting a child who chose a vegetarian sandwich is $\frac{9}{50}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

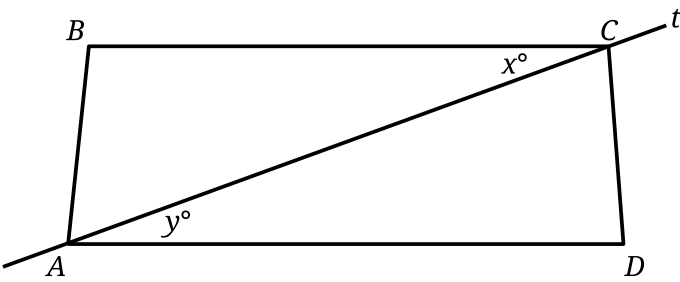
Question Difficulty:

Easy

Question ID 9342ca50

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 9342ca50



Note: Figure not drawn to scale.

In the figure shown, $\overline{BC}$ is parallel to $\overline{AD}$ and $x = 22$. Line t passes through points A and C . What is the value of y ?

- A. 22
- B. 44
- C. 68
- D. 158

ID: 9342ca50 Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that $\overline{BC}$ is parallel to $\overline{AD}$. Since line t intersects $\overline{BC}$ and $\overline{AD}$, it's a transversal. The angles in the figure marked as x° and y° are on opposite sides of the transversal and lie between the parallel segments, so they are alternate interior angles. When two parallel segments are intersected by a transversal, alternate interior angles have equal measure. It's also given that $x = 22$. Therefore, the value of y is 22.

Choice B is incorrect. This is the value of $2x$, not the value of y .

Choice C is incorrect. This is the complement of 22, not the value of y .

Choice D is incorrect. This is the supplement of 22, not the value of y .

Question Difficulty:
Easy

Question ID 96467fea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 96467fea

Circle N has a radius of **7** millimeters (**mm**). Circle M has an area of **64π mm²**. What is the total area, in **mm²**, of circles N and M?

- A. **113π**
- B. **92π**
- C. **78π**
- D. **15π**

ID: 96467fea Answer

Correct Answer:
A

Rationale

Choice A is correct. It's given that circle N has a radius of **7 mm**. The area of a circle is given by the expression **πr^2** , where **r** is the radius of the circle. Substituting **7** for **r** in the expression yields **$\pi(7)^2$** , or **49π** , for the area, in **mm²**, of circle N. It's also given that the area of circle M is **64π mm²**. Adding the two areas yields **$49\pi + 64\pi$** , or **113π , mm²**. Therefore, the total area, in **mm²**, of circles N and M is **113π** .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Medium

Question ID 6b4707aa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 6b4707aa

Circle A in the xy -plane has the equation $(x + 5)^2 + (y - 5)^2 = 25$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = k$, where k is a constant. What is the value of k ?

ID: 6b4707aa Answer

Correct Answer:
100

Rationale

The correct answer is **100**. An equation of a circle in the xy -plane can be written as $(x - t)^2 + (y - u)^2 = r^2$, where the center of the circle is (t, u) , the radius of the circle is r , and where t , u , and r are constants. It's given that the equation of circle A is $(x + 5)^2 + (y - 5)^2 = 25$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 5^2$. Therefore, the center of circle A is $(-5, 5)$ and the radius of circle A is **5**. It's given that circle B has the same center as circle A and that the radius of circle B is two times the radius of circle A. Therefore, the center of circle B is $(-5, 5)$ and the radius of circle B is **2(5)**, or **10**. Substituting -5 for t , **5** for u , and **10** for r in the equation $(x - t)^2 + (y - u)^2 = r^2$ yields $(x + 5)^2 + (y - 5)^2 = 10^2$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 100$. It follows that the equation of circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = 100$. It's also given that the equation defining circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = k$, where k is a constant. Therefore, the value of k is **100**.

Question Difficulty:
Hard

Question ID c447367f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: c447367f

A circle in the xy -plane has its center at $(19, 18)$ and has a radius of $9k$. Which equation represents this circle?

- A. $(x - 19)^2 + (y - 18)^2 = 81k$
- B. $(x - 19)^2 + (y - 18)^2 = 81k^2$
- C. $(x - 19)^2 + (y - 18)^2 = 9k$
- D. $(x - 19)^2 + (y - 18)^2 = 9k^2$

ID: c447367f Answer

Correct Answer:
B

Rationale

Choice B is correct. The equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r . It's given that this circle has a center at $(19, 18)$ and a radius of $9k$. Substituting 19 for h , 18 for k , and $9k$ for r in $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 19)^2 + (y - 18)^2 = (9k)^2$, or $(x - 19)^2 + (y - 18)^2 = 81k^2$. Therefore, the equation that represents this circle is $(x - 19)^2 + (y - 18)^2 = 81k^2$.

Choice A is incorrect. This equation represents a circle with radius $9\sqrt{k}$, not $9k$.

Choice C is incorrect. This equation represents a circle with radius $\sqrt{9k}$, not $9k$.

Choice D is incorrect. This equation represents a circle with radius $\sqrt{9k}$, not $9k$.

Question Difficulty:
Hard

Question ID 5011b039

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	

ID: 5011b039

What is the radius of the circle in the xy -plane defined by $(x + 2)^2 + (y + 5)^2 = 169$?

ID: 5011b039 Answer

Correct Answer:
13

Rationale

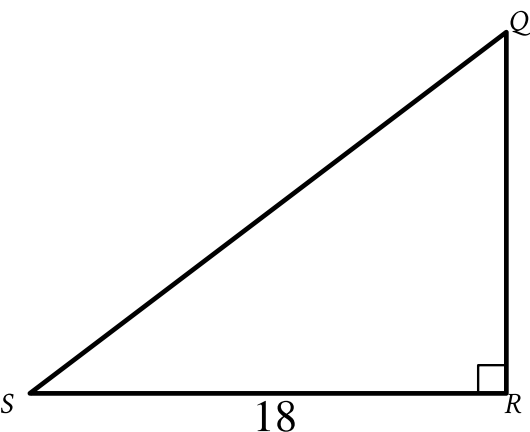
The correct answer is **13**. The equation of a circle in the xy -plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and r is the radius of the circle. It's given that the circle in the xy -plane is defined by $(x + 2)^2 + (y + 5)^2 = 169$. Therefore, $r^2 = 169$. Taking the positive square root of both sides of this equation yields $r = 13$. Thus, the radius of the circle is **13**.

Question Difficulty:
Medium

Question ID 94f7c9e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 94f7c9e2



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of QS ?

- A. $18 \cos Q$
- B. $18 \sin Q$
- C. $\frac{18}{\cos Q}$
- D. $\frac{18}{\sin Q}$

ID: 94f7c9e2 Answer

Correct Answer:
D

Rationale

Choice D is correct. The figure shows that triangle QRS is a right triangle. Each of the given choices is an expression containing $\sin Q$ or $\cos Q$. For an acute angle in a right triangle, the sine of the angle is the length of the opposite leg divided by the length of the hypotenuse, and the cosine of the angle is the length of the adjacent leg divided by the length of the hypotenuse. It follows that $\sin Q = \frac{RS}{QS}$ and $\cos Q = \frac{QR}{QS}$, where $QR < RS$. It's given in the figure that the length of side RS is 18. Since only the equation $\sin Q = \frac{RS}{QS}$ contains RS , an expression representing QS can be found by substituting 18 for RS in this equation, which yields $\sin Q = \frac{18}{QS}$. Multiplying both sides of this equation by QS yields $QS \cdot \sin Q = 18$. Dividing both sides of this equation by $\sin Q$ yields $QS = \frac{18}{\sin Q}$. Therefore, the expression $\frac{18}{\sin Q}$ represents the length of QS .

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Hard

Question ID 62ec6574

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 62ec6574

A right rectangular prism has a base area of $24t$ square centimeters (cm^2). The length of the base of the rectangular prism is $\frac{8}{3}$ cm, and the height of the rectangular prism is 15 cm. Which expression represents the surface area, in cm^2 , of the right rectangular prism?

- A. $48t + 160$
- B. $318t + 80$
- C. $1,968t + 80$
- D. $360t$

ID: 62ec6574 Answer

Correct Answer:
B

Rationale

Choice B is correct. It's given that the base area of the right rectangular prism is $24t \text{ cm}^2$ and the length of the base is $\frac{8}{3}$ cm. Dividing the base area by the length yields the width: $\frac{24t}{\frac{8}{3}}$, or $(24t)(\frac{3}{8})$, or $9t$ cm. It's also given that the height of the prism is 15 cm. A right rectangular prism has two rectangular bases and four rectangular lateral faces. The total area of the two bases is $2(24t)$, or $48t \text{ cm}^2$. The four lateral faces include two with dimensions $\frac{8}{3}$ cm by 15 cm and two with dimensions $9t$ cm by 15 cm. The total area of these four lateral faces is $2(\frac{8}{3})(15) + 2(9t)(15) \text{ cm}^2$, which is equivalent to $80 + 270t \text{ cm}^2$. Adding this to the total area of the two bases yields $48t + 270t + 80$, or $318t + 80 \text{ cm}^2$. Thus, the expression, which represents the surface area, in cm^2 , of the right rectangular prism is $318t + 80$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the volume, in cm^3 , not the surface area, in cm^2 .

Question Difficulty:
Hard

Question ID c41eb616

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: c41eb616

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 140)^\circ$. Which of the following could **NOT** be the sum of the measures of any two of these angles?

- A. $(-18x + 280)^\circ$
- B. $(-18x + 640)^\circ$
- C. $(18x - 280)^\circ$
- D. 180°

ID: c41eb616 Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that two lines intersect at exactly one point, forming two acute angles and two obtuse angles. When two lines intersect, opposite angles are congruent and adjacent angles are supplementary. It's given that the measure of one of the angles is $(9x - 140)^\circ$, so the measure of the opposite angle is $(9x - 140)^\circ$, and the measure of each adjacent angle is $(180 - (9x - 140))^\circ$, or $(-9x + 320)^\circ$. The possible sums of the measures of any two of the angles are $(9x - 140)^\circ + (9x - 140)^\circ$, or $(18x - 280)^\circ$, $(-9x + 320)^\circ + (-9x + 320)^\circ$, or $(-18x + 640)^\circ$, and $(9x - 140)^\circ + (-9x + 320)^\circ$, or 180° . Of the given choices, only $(-18x + 280)^\circ$ is not equivalent to any of these expressions and could not be the sum of the measures of any two of the angles.

Choice B is incorrect. This is the sum of the measures of two angles that are adjacent to the given angle.

Choice C is incorrect. This is the sum of the measure of the given angle and the measure of its opposite angle.

Choice D is incorrect. This is the sum of the measure of the given angle and the measure of an angle adjacent to the given angle.

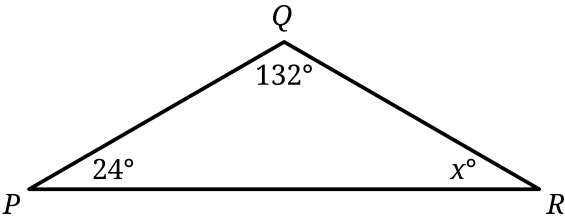
Question Difficulty:

Hard

Question ID 7268586c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 7268586c



Note: Figure not drawn to scale.

In the triangle shown, $PQ = QR$. What is the value of x ?

- A. 156
- B. 66
- C. 48
- D. 24

ID: 7268586c Answer

Correct Answer:
D

Rationale

Choice D is correct. The sum of the angle measures of a triangle is 180° . It's given that the measure of angle Q is 132° , the measure of angle P is 24° , and the measure of angle R is x° . It follows that $x + 132 + 24 = 180$, or $x + 152 = 180$. Subtracting 152 from both sides of this equation yields $x = 24$.

Alternate approach: It's given that in the triangle shown, $PQ = QR$. Thus, triangle PQR is an isosceles triangle. Since base angles of an isosceles triangle are congruent, the measure of angle P is equal to the measure of angle R . Therefore, the measure of angle R is 24° , and the value of x is 24 .

Choice A is incorrect. This is the sum of the measures, in degrees, of angles Q and P , not the measure, in degrees, of angle R .

Choice B is incorrect and may result from conceptual or calculation errors.

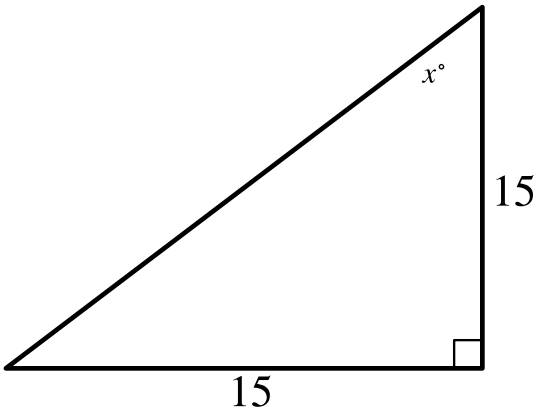
Choice C is incorrect. This is the sum of the measures, in degrees, of angles P and R , not the measure, in degrees, of angle R .

Question Difficulty:
Easy

Question ID 9823f279

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 9823f279



Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

ID: 9823f279 Answer

Correct Answer:
45

Rationale

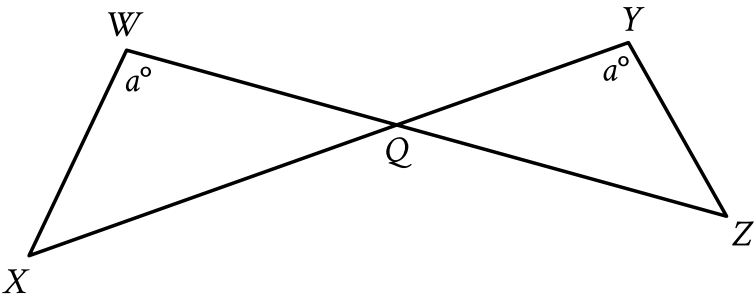
The correct answer is **45**. An isosceles right triangle has a right angle and two legs of equal length. In the triangle shown, one angle is a right angle and the two legs each have a length of **15**. Thus, the given triangle is an isosceles right triangle. In an isosceles right triangle, the measures of the two non-right angles are **45°** . It follows that the value of x is **45**.

Question Difficulty:
Medium

Question ID 1318dfea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 1318dfea



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q , $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

- A. 18
- B. 36
- C. 120
- D. 200

ID: 1318dfea Answer

Correct Answer:

A

Rationale

Choice A is correct. It's given that $\overline{WZ}$ and $\overline{XY}$ intersect at point Q . It follows that $\angle WQX$ and $\angle YQZ$ are vertical angles, which means they are congruent to each other. The figure shows that $\angle W$ and $\angle Y$ both have a measure of a° , so they are congruent to each other. Therefore, $\triangle QWX$ and $\triangle QYZ$ are similar triangles, where $\overline{WQ}$ corresponds to $\overline{YQ}$ and $\overline{WX}$ corresponds to $\overline{YZ}$. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{YZ}{YQ} = \frac{WX}{WQ}$. It's given that $YQ = 21$, $WQ = 70$, and $WX = 60$. Substituting 21 for YQ , 70 for WQ , and 60 for WX in the equation $\frac{YZ}{YQ} = \frac{WX}{WQ}$ yields $\frac{YZ}{21} = \frac{60}{70}$, or $\frac{YZ}{21} = \frac{6}{7}$. Multiplying each side of this equation by 21 yields $YZ = 18$. Therefore, the length of $\overline{YZ}$ is 18.

Choice B is incorrect. This is the length of $\overline{QZ}$, not $\overline{YZ}$.

Choice C is incorrect. This is the length of $\overline{XQ}$, not $\overline{YZ}$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Hard

Question ID c4e5e23c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: c4e5e23c

The length of one side of square M is 6 times the length of one side of square N. The area of square N is 49 square feet. What is the area, in square feet, of square M?

- A. 1,764
- B. 504
- C. 294
- D. 252

ID: c4e5e23c Answer

Correct Answer:

A

Rationale

Choice A is correct. The area, A , of a square is given by the equation $A = s^2$, where s is the length of one side of the square. Let x represent the length, in feet, of one side of square N. It's given that the area of square N is 49 square feet; therefore, it follows that $49 = x^2$. It's also given that the length of one side of square M is 6 times the length of one side of square N. Therefore, the length of one side of square M can be represented by the expression $6x$. Substituting $6x$ for s in the equation $A = s^2$ yields $A = (6x)^2$, or $A = 36x^2$. Since $49 = x^2$, substituting 49 for x^2 in the equation $A = 36x^2$ yields $A = 36(49)$, or $A = 1,764$. Therefore, the area, in square feet, of square M is 1,764.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question Difficulty:

Medium

Question ID 585a9138

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	

ID: 585a9138

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 27° . What is the measure, in degrees, of angle F ?

ID: 585a9138 Answer

Correct Answer:
63

Rationale

The correct answer is **63**. It's given that triangles ABC and DEF are congruent with angle A corresponding to angle D . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that the measure of angle A is 27° . It follows that the measure of angle D is also 27° . It's given that angle E is a right angle. Therefore, the measure of angle E is 90° . Let x represent the measure, in degrees, of angle F . Since the measures of the angles in a triangle sum to 180° , it follows that $27 + 90 + x = 180$, or $117 + x = 180$. Subtracting **117** from both sides of this equation yields $x = \mathbf{63}$. Therefore, the measure, in degrees, of angle F is **63**.

Question Difficulty:
Medium

Question ID 603c18ad

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	

ID: 603c18ad

In triangle XYZ , the measure of angle X is 90° . Point W lies on segment YZ , and segment WX is perpendicular to segment YZ . The length of segment WY is 572 , and the length of segment WX is 429 . What is the value of $\tan Z$?

- A. $\frac{3}{5}$
- B. $\frac{3}{4}$
- C. $\frac{4}{5}$
- D. $\frac{4}{3}$

ID: 603c18ad Answer

Correct Answer:

D

Rationale

Choice D is correct. It's given that in triangle XYZ , the measure of angle X is 90° , so triangle XYZ is a right triangle. It's also given that point W lies on segment YZ , and segment WX is perpendicular to segment YZ . It follows that triangle WYX is a right triangle. Triangle WYX and triangle XYZ are right triangles with the same interior angle measure at point Y . It follows that triangle WYX and triangle XYZ are similar, where angles W , Y , and X in triangle WYX correspond to angles X , Y , and Z , respectively, in triangle XYZ . Since corresponding angles in similar triangles are congruent, their tangents have equal value. Thus, the tangent of angle Z is equal to the tangent of angle X in triangle WYX . The opposite side of angle X in triangle WYX is segment WY , and the adjacent side is segment WX . It's given that the length of segment WY is 572 and the length of segment WX is 429 . It follows that the tangent of angle X in triangle WYX is $\frac{572}{429}$, or $\frac{4}{3}$. Therefore, the value of $\tan Z$ is $\frac{4}{3}$.

Choice A is incorrect. This is the value of $\cos Z$, not $\tan Z$.

Choice B is incorrect. This is the value of $\tan Y$, not $\tan Z$.

Choice C is incorrect. This is the value of $\sin Z$, not $\tan Z$.

Question Difficulty:

Hard

Question ID 9d2e7037

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	

ID: 9d2e7037

A rectangle has a length of **56** inches and a width of **28** inches. What is the area, in square inches, of the rectangle?

- A. **28**
- B. **84**
- C. **168**
- D. **1,568**

ID: 9d2e7037 Answer

Correct Answer:
D

Rationale

Choice D is correct. The area, ***A***, of a rectangle can be found using the formula **$A = \ell w$** , where ***ℓ*** represents the length and ***w*** represents the width of the rectangle. It's given that a rectangle has a length of **56** inches and a width of **28** inches. Substituting **56** for ***ℓ*** and **28** for ***w*** in the formula **$A = \ell w$** yields **$A = (56)(28)$** , or **$A = 1,568$** . Therefore, the area, in square inches, of the rectangle is **1,568**.

Choice A is incorrect. This is the width, in inches, not the area, in square inches, of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question Difficulty:
Easy